PCA - Erdbeere Gesamt v2 · Without Threshold · Without OdourType · Iterative Outlier Removal

Version: v2 – no OAV / olfactory threshold, no OdourType grouping. **Goal:** PCA purely on aroma substances (CAS) and normalised recipe proportions. **Pipeline:** Remove ignore-list substances (CAS-based) → normalise Totalmenge per recipe → **Iterative outlier removal** (max 20 iterations): detect outlier(s) in PC1/PC2 space, note them, remove them, repeat until no outliers remain or max iterations reached. At each iteration a plot is produced and outlier recipe IDs are recorded.

Dataset: `data/gold/Third_Trial_Set_PDM Erdbeere Gesamt 8-5-2026.csv`
Outputs: `outputs/pca_v2_noOT_*`

In [1]:
```
import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.lines import Line2D
import matplotlib.cm as cm
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from pathlib import Path

# ── paths ──────────────────────────────────────────────────────────────────
BASE      = Path('..')
DATA_PATH = BASE / 'data/gold/Third_Trial_Set_PDM Erdbeere Gesamt 8-5-2026.c
OUT_DIR   = BASE / 'outputs'
OUT_DIR.mkdir(exist_ok=True)
IGNORE_PATH = BASE / 'data/gold/ignone_substances.csv'

print('Libraries loaded ✓')
```

```
Libraries loaded ✓
```

1 · Load & Clean Data

In [2]:
```
def parse_de_float(val):
    if pd.isna(val):
        return float('nan')
    s = str(val).strip().replace(',', '.')
    try:
        return float(s)
    except ValueError:
```

```
        return float('nan')

df = pd.read_csv(DATA_PATH, dtype=str)
df['Totalmenge'] = df['Totalmenge'].apply(parse_de_float)
df['Threshold']  = df['Threshold'].apply(parse_de_float)

print(f'Raw rows: {len(df):,}')
print(f'Unique recipes (Rez.-Nr.): {df["Rez.-Nr."].nunique()}')
print(f'Unique ingredients (Ident): {df["Ident"].nunique()}')
print(f'Unique CAS numbers: {df["CAS-Nr."].nunique()}')
```

```
Raw rows: 3,369
Unique recipes (Rez.-Nr.): 130
Unique ingredients (Ident): 338
Unique CAS numbers: 231
```

In [3]:
```
# ── Apply ignore list ──────────────────────────────────────────────
if IGNORE_PATH.exists():
    ign             = pd.read_csv(IGNORE_PATH)
    ign_idents      = set(ign['Ident'].dropna().astype(str).str.strip())
    names_to_ignore = {str(n).lower().strip() for n in ign['Name']}
    mask = (
        df['Ident'].astype(str).str.strip().isin(ign_idents) |
        df['Name'].str.lower().str.strip().isin(names_to_ignore)
    )
    cas_to_ignore = set(df.loc[mask, 'CAS-Nr.'].dropna().astype(str).str.str
    df.loc[df['CAS-Nr.'].astype(str).str.strip().isin(cas_to_ignore), 'Total
    print(f'Ignored idents: {len(ign_idents)} | Ignored CAS numbers: {len(ca
else:
    print('No ignore list found — all ingredients included.')
```

```
Ignored idents: 10 | Ignored CAS numbers: 6
```

In [4]:
```
# -- Normalize Totalmenge per recipe (no threshold, no OAV) ----------------
df_t = df[df['Totalmenge'] > 0].copy()

per_recipe_sum = df_t.groupby('Rez.-Nr.')['Totalmenge'].transform('sum')
df_t['Norm_Totalmenge'] = df_t['Totalmenge'] / per_recipe_sum

print(f'Rows with positive Totalmenge: {len(df_t):,}')
print(f'Recipes after filtering      : {df_t["Rez.-Nr."].nunique()}')
print(f'Unique CAS numbers retained  : {df_t["CAS-Nr."].nunique()}')
print()
print('Norm_Totalmenge distribution:')
print(df_t['Norm_Totalmenge'].describe().round(5))
```

```
Rows with positive Totalmenge: 3,155
Recipes after filtering      : 130
Unique CAS numbers retained  : 225

Norm_Totalmenge distribution:
count    3155.00000
mean        0.04120
std         0.10592
min         0.00000
25%         0.00093
50%         0.00661
75%         0.03654
max         1.00000
Name: Norm_Totalmenge, dtype: float64
```

In [5]:
```
# -- Build Recipe x CAS Normalized Totalmenge matrix ----------------------
pivot_oav = df_t.pivot_table(
    index='Rez.-Nr.', columns='CAS-Nr.',
    values='Norm_Totalmenge', aggfunc='sum', fill_value=0
)
print(f'Recipe x CAS matrix       : {pivot_oav.shape}')
print(f'Avg CAS per recipe        : {(pivot_oav > 0).sum(axis=1).mean():.1f}
print(f'Matrix value range        : {pivot_oav.values.min():.5f} - {pivot_oa

# No log transformation — normalised proportions are already on a comparable
X_oav = pivot_oav.values.copy()
```

```
Recipe x CAS matrix       : (129, 225)
Avg CAS per recipe        : 23.8
Matrix value range        : 0.00000 - 0.78706
```

In [6]:
```
# -- Ingredient Statistics: Frequency & Average (after removal + normalisati
cas_name_full = df.groupby('CAS-Nr.')['Name'].first().to_dict()

freq      = (pivot_oav > 0).mean(axis=0)
n_recipes = (pivot_oav > 0).sum(axis=0)

avg_present_mat = pivot_oav.copy()
avg_present_mat[avg_present_mat == 0] = np.nan
avg_when_present = avg_present_mat.mean(axis=0)
avg_all = pivot_oav.mean(axis=0)

ing_stats = pd.DataFrame({
    'CAS':                  pivot_oav.columns,
    'Ingredient':           [cas_name_full.get(c, c) for c in pivot_oav.col
    'Frequency':            freq.values.round(4),
    'Recipes_Count':        n_recipes.values,
    'Avg_Norm_When_Present': avg_when_present.values.round(6),
    'Avg_Norm_All_Recipes':  avg_all.values.round(6),
}).sort_values('Frequency', ascending=False).reset_index(drop=True)

print(f'Ingredient statistics for {len(ing_stats)} unique CAS numbers '
      f'across {len(pivot_oav)} recipes')
print()
print(ing_stats.head(30).to_string(index=False))
```

```
ing_stats.to_csv(OUT_DIR / 'pca_v2_noOT_ingredient_statistics.csv', index=Fa
ing_stats.to_excel(OUT_DIR / 'pca_v2_noOT_ingredient_statistics.xlsx', index
print('\nExported: pca_v2_noOT_ingredient_statistics.csv / .xlsx')
```

Ingredient statistics for 225 unique CAS numbers across 129 recipes

CAS	Ingredient	Frequency	Recipes_Count	Avg_Norm_When_Present	Avg_Norm_All_Recipes
105-54-4	Ethylbutyrat Kosher Halal	0.9922	128	0.093956	0.093228
706-14-9	gamma-Decalacton Kosher Halal	0.9302	120	0.038739	0.036037
123-66-0	Ethylhexanoat Halal	0.9225	119	0.036157	0.033354
7452-79-1	Ethyl-2-methylbutyrat Kosher Halal	0.9147	118	0.048705	0.044552
928-96-1	cis-3-Hexen-1-ol Halal Kosher	0.8992	116	0.088893	0.079935
116-53-0	2-Methylbuttersäure Halal Kosher	0.8140	105	0.063624	0.051787
3658-77-3	Furaneol Halal	0.7984	103	0.095135	0.075961
142-62-1	Hexansäure kräftig	0.7829	101	0.023316	0.018255
3681-71-8	cis-3-Hexenylacetat Halal Kosher	0.7442	96	0.011774	0.008762
141-78-6	Ethylacetat Kosher Halal	0.6124	79	0.033561	0.020553
64-19-7	Essigsäure Halal Kosher	0.5969	77	0.036827	0.021982
108-64-5	Ethylisovalerat Kosher Halal	0.5814	75	0.015013	0.008728
107-92-6	Buttersäure Halal Kosher	0.5659	73	0.021253	0.012027
142-92-7	Hexylacetat Kosher Halal	0.5116	66	0.008638	0.004419
118-71-8	Maltol Kosher Halal	0.5116	66	0.046309	0.023693
121-33-5	Vanillin ex Eugenol, natürlich Halal Kosher, BG	0.4884	63	0.031729	0.015495
111-27-3	Hexanol Kosher Halal	0.4574	59	0.014148	0.006471
103-26-4	Methyl-trans-cinnamat, natürlich Kosher Halal	0.4341	56	0.034807	0.015110
1754-62-7	Methyl-trans-cinnamat Halal	0.4264	55	0.055850	0.023812
659-70-1	Isoamylisovalerat Kosher Halal	0.3798	49	0.011154	0.004237
16491-36-4	cis-3-Hexenylbutyrat Halal	0.3023	39	0.006133	0.001854
140-11-4	Benzylacetat Halal	0.2946	38	0.004795	0.001412
75-07-0	Acetaldehyd Halal Kosher	0.2636	34	0.006845	0.001804
123-51-3	Isoamylalkohol Kosher Halal	0.2636	34	0.009783	0.002578
78-70-6	Linalool ex HO, natürlich Kosher Halal BG	0.2636	34	0.002670	0.000704
513-86-0	Acetoin 50% in PG Kosher Halal	0.2558	33	0.040827	0.010444

```
  100-52-7                        Benzaldehyd Kosher Halal     0.2558
33               0.001532              0.000392
  123-92-2                         Isoamylacetat Halal     0.2171
28               0.026714              0.005798
31501-11-8                   cis-3-Hexenylhexanoat Halal     0.2093
27               0.011465              0.002400
  106-27-4                        Isoamylbutyrat Halal     0.1938
25               0.009425              0.001827

Exported: pca_v2_noOT_ingredient_statistics.csv / .xlsx
```

2 · Iterative PCA – Outlier Detection & Removal

Method: At each iteration:

1. Fit PCA on the current recipe set.
2. Compute z-scores of PC1 and PC2 scores across all active recipes.
3. Compute 2D Mahalanobis distance: `d = sqrt(z1² + z2²)`.
4. Flag recipes with `d > Z_THRESH` as outliers (capped at `MAX_REMOVE_PER_IT = 3` per iteration, worst-first).
5. Plot all recipes; outliers are marked in red with recipe ID labels.
6. Remove outliers → repeat.
7. Stop when no outliers remain or after `MAX_ITER = 20` iterations.

The **final clean dataset** is used for all subsequent analyses.

What does the 4.5σ threshold mean?

Step by step:

1. After PCA, every recipe gets a score on PC1 and PC2.
2. Those scores are standardised to z-scores:
   - `z1 = (PC1_score − mean(PC1)) / std(PC1)`
   - `z2 = (PC2_score − mean(PC2)) / std(PC2)`
3. A single **distance from the centroid** is computed:
   - `d = sqrt(z1² + z2²)`
4. Any recipe with `d > 4.5` is flagged as an outlier.

Geometrically: this is the radius of an ellipse in PC1–PC2 space centred on the average recipe. The ellipse's axes are `4.5 × std(PC1)` and `4.5 × std(PC2)`. Recipes **outside** this ellipse are removed.

What it is NOT: 4.5 standard deviations above/below the mean on a single axis. A recipe at 3σ on PC1 *and* 3σ on PC2 has a combined distance of `sqrt(9+9) ≈ 4.24σ` — inside the boundary. Only recipes extreme in the **combined 2D sense** are removed.

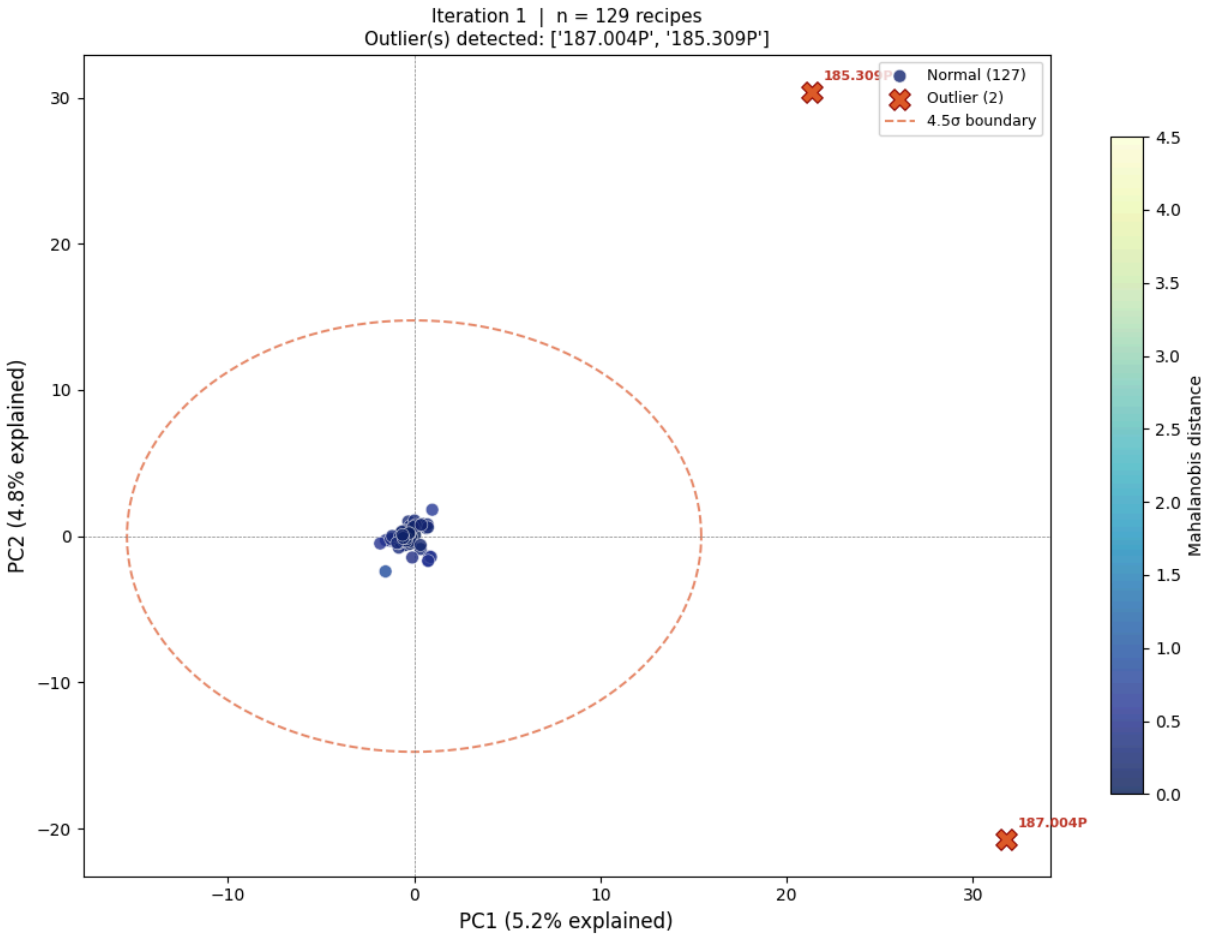
Statistical calibration: Under a bivariate normal distribution, the probability of a point falling beyond radius 4.5 is `1 − χ²_cdf(4.5², df=2) ≈ 0.004%`. With 129 recipes, essentially zero false positives are expected — so any recipe flagged is genuinely compositionally extreme.

In [7]:

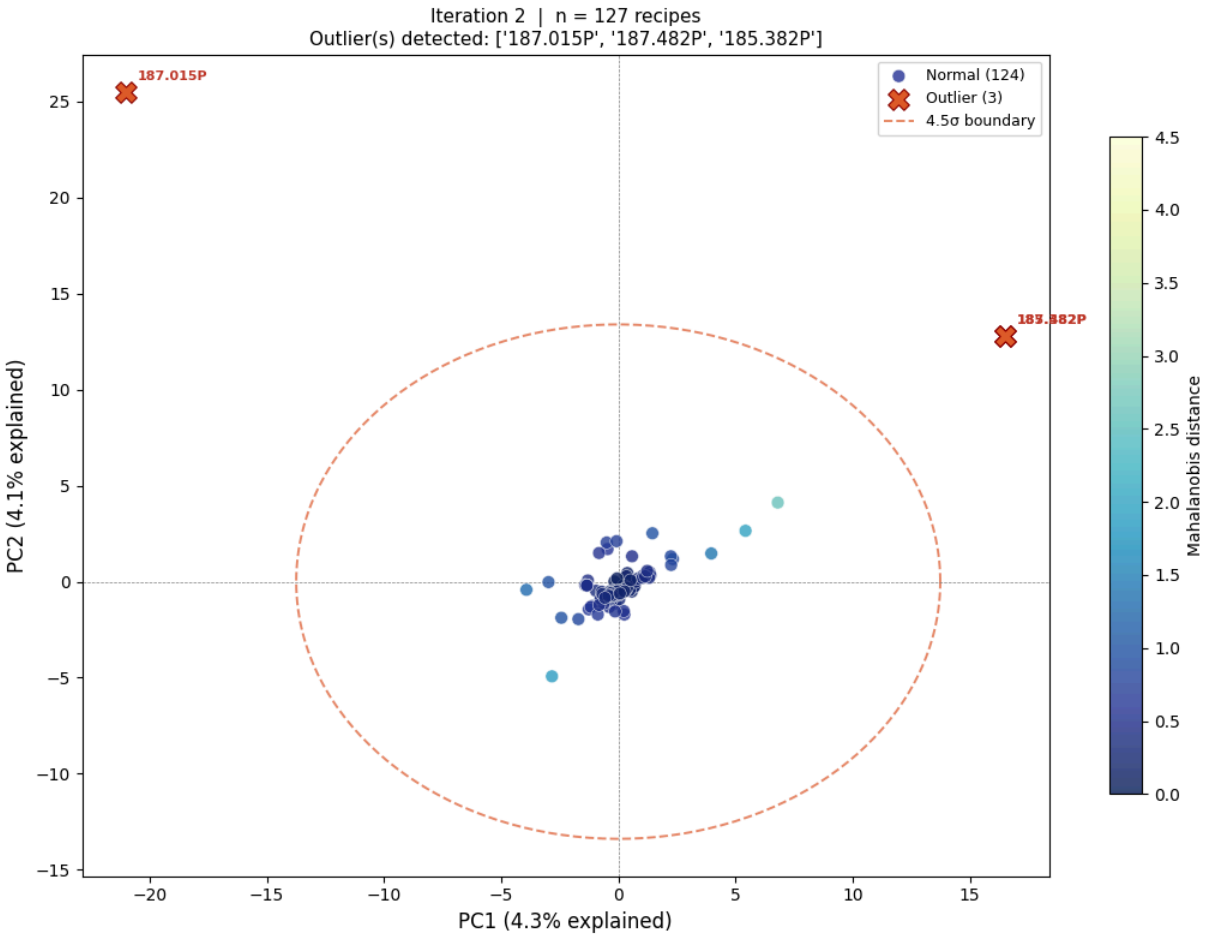
```
MAX_ITER          = 20
Z_THRESH          = 4.5   # 2D Mahalanobis distance threshold (sigma)
MAX_REMOVE_PER_IT = 3     # max outliers removed per iteration

# Working copies
active_recipes = list(pivot_oav.index)
active_X       = X_oav.copy()

all_removed    = []   # [(iteration, rez_nr), ...]
iteration_log  = []   # metadata per iteration

# Will be set when the loop ends
final_scaler   = None
final_pca      = None
final_scores   = None
final_ev       = None
final_active_recipes = None
final_X_sc     = None

for it in range(1, MAX_ITER + 1):
    # ── Scale ──────────────────────────────────────────────────────────
    scaler = StandardScaler()
    X_sc   = scaler.fit_transform(active_X)

    # ── PCA ────────────────────────────────────────────────────────────
    n_comp = min(10, X_sc.shape[1], X_sc.shape[0] - 1)
    pca    = PCA(n_components=n_comp, random_state=42)
    scores = pca.fit_transform(X_sc)
    ev     = pca.explained_variance_ratio_ * 100

    # ── Outlier detection: 2D Mahalanobis in PC1–PC2 space ─────────────
    s1 = scores[:, 0]
    s2 = scores[:, 1]
    z1 = (s1 - s1.mean()) / (s1.std() + 1e-12)
    z2 = (s2 - s2.mean()) / (s2.std() + 1e-12)
    mahal        = np.sqrt(z1**2 + z2**2)
    # Keep at most MAX_REMOVE_PER_IT extreme outliers per iteration
    cand_idx     = np.where(mahal > Z_THRESH)[0]
    # Sort candidates by distance (worst first) and cap
    cand_sorted = cand_idx[np.argsort(mahal[cand_idx])[::-1]][:MAX_REMOVE_PE
    outlier_idx = cand_sorted
    outlier_mask = np.zeros(len(active_recipes), dtype=bool)
    outlier_mask[outlier_idx] = True
    outlier_rez  = [active_recipes[i] for i in outlier_idx]

    # ── 3-sigma ellipse in score space ─────────────────────────────────
    theta   = np.linspace(0, 2 * np.pi, 300)
    ell_cx, ell_cy = s1.mean(), s2.mean()
    ell_ax, ell_ay = Z_THRESH * s1.std(), Z_THRESH * s2.std()
```

```
ell_x = ell_cx + ell_ax * np.cos(theta)
ell_y = ell_cy + ell_ay * np.sin(theta)

# ── Plot ──────────────────────────────────────────────────────────────
fig, ax = plt.subplots(figsize=(11, 8))

# Color normal points by Mahalanobis distance (cool=close, warm=far)
cmap   = plt.get_cmap('Blues')
norm_d = plt.Normalize(vmin=0, vmax=Z_THRESH)
non_out = ~outlier_mask
sc = ax.scatter(
    s1[non_out], s2[non_out],
    c=mahal[non_out], cmap='YlGnBu_r', norm=norm_d,
    s=60, alpha=0.80, edgecolors='white', linewidths=0.4,
    label=f'Normal ({non_out.sum()})', zorder=3
)
plt.colorbar(sc, ax=ax, label='Mahalanobis distance', shrink=0.8)

if len(outlier_idx):
    ax.scatter(
        s1[outlier_mask], s2[outlier_mask],
        c='#E05A2B', s=160, alpha=1.0, marker='X', zorder=5,
        edgecolors='#8B0000', linewidths=0.8,
        label=f'Outlier ({outlier_mask.sum()})'
    )
    for i in outlier_idx:
        ax.annotate(
            str(active_recipes[i]),
            (s1[i], s2[i]),
            xytext=(7, 7), textcoords='offset points',
            fontsize=8, color='#C0392B', fontweight='bold'
        )

ax.plot(ell_x, ell_y, '--', color='#E05A2B', lw=1.4, alpha=0.7,
        label=f'{Z_THRESH:.1f}σ boundary')
ax.axhline(0, color='grey', lw=0.5, ls='--')
ax.axvline(0, color='grey', lw=0.5, ls='--')
ax.set_xlabel(f'PC1 ({ev[0]:.1f}% explained)', fontsize=12)
ax.set_ylabel(f'PC2 ({ev[1]:.1f}% explained)', fontsize=12)

if outlier_rez:
    status = f'Outlier(s) detected: {outlier_rez}'
else:
    status = 'No outliers — convergence reached ✓'

ax.set_title(
    f'Iteration {it}  |  n = {len(active_recipes)} recipes\n{status}',
    fontsize=11
)
ax.legend(loc='upper right', fontsize=9)
plt.tight_layout()
fname = f'pca_v2_noOT_iter{it:02d}.png'
fig.savefig(OUT_DIR / fname, dpi=150, bbox_inches='tight')
plt.show()
print(f'Saved: {fname}')
```

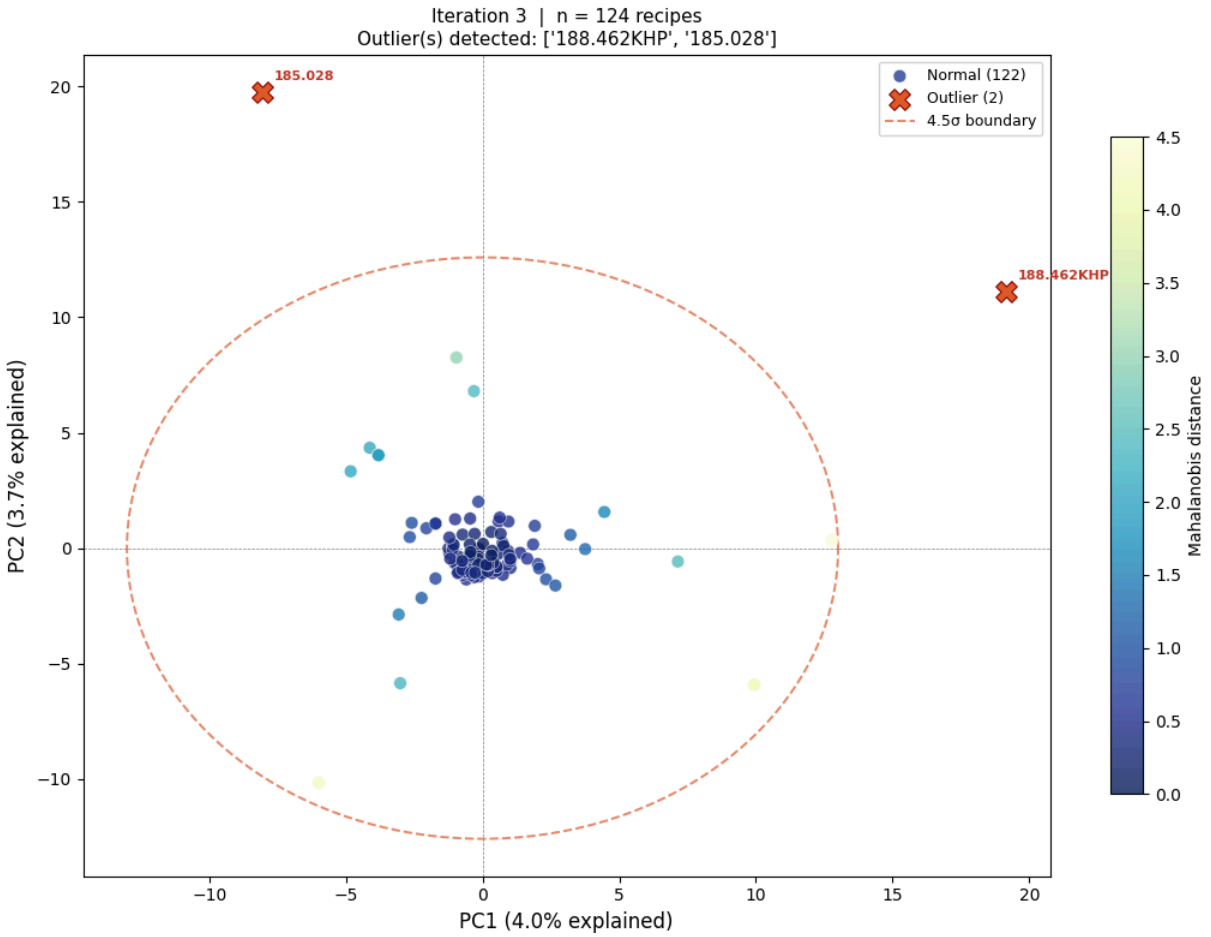
```
    iteration_log.append({
        'iteration': it,
        'n_recipes': len(active_recipes),
        'outliers':  outlier_rez,
        'n_outliers': len(outlier_rez),
        'ev_PC1':    round(ev[0], 1),
        'ev_PC2':    round(ev[1], 1),
        'ev_cum3':   round(sum(ev[:3]), 1),
    })

    # ── Store final-state variables ──────────────────────────────────────
    final_scaler         = scaler
    final_pca            = pca
    final_scores         = scores
    final_ev             = ev
    final_active_recipes = active_recipes.copy()
    final_X_sc           = X_sc

    if not outlier_rez:
        print(f'\n✓ Iteration {it}: No outliers — converged.')
        print(f'  Final clean dataset: {len(active_recipes)} recipes')
        break

    print(f'Iteration {it}: removing {len(outlier_rez)} outlier(s): {outlier
    for r in outlier_rez:
        all_removed.append((it, r))

    keep           = ~outlier_mask
    active_recipes = [r for r, k in zip(active_recipes, keep) if k]
    active_X       = active_X[keep]

else:
    print(f'\n⚠ Maximum {MAX_ITER} iterations reached — stopping.')
    print(f'  Remaining dataset: {len(active_recipes)} recipes')

print(f'\nTotal removed: {len(all_removed)} recipe(s)')
print('Removed list:', all_removed)
```



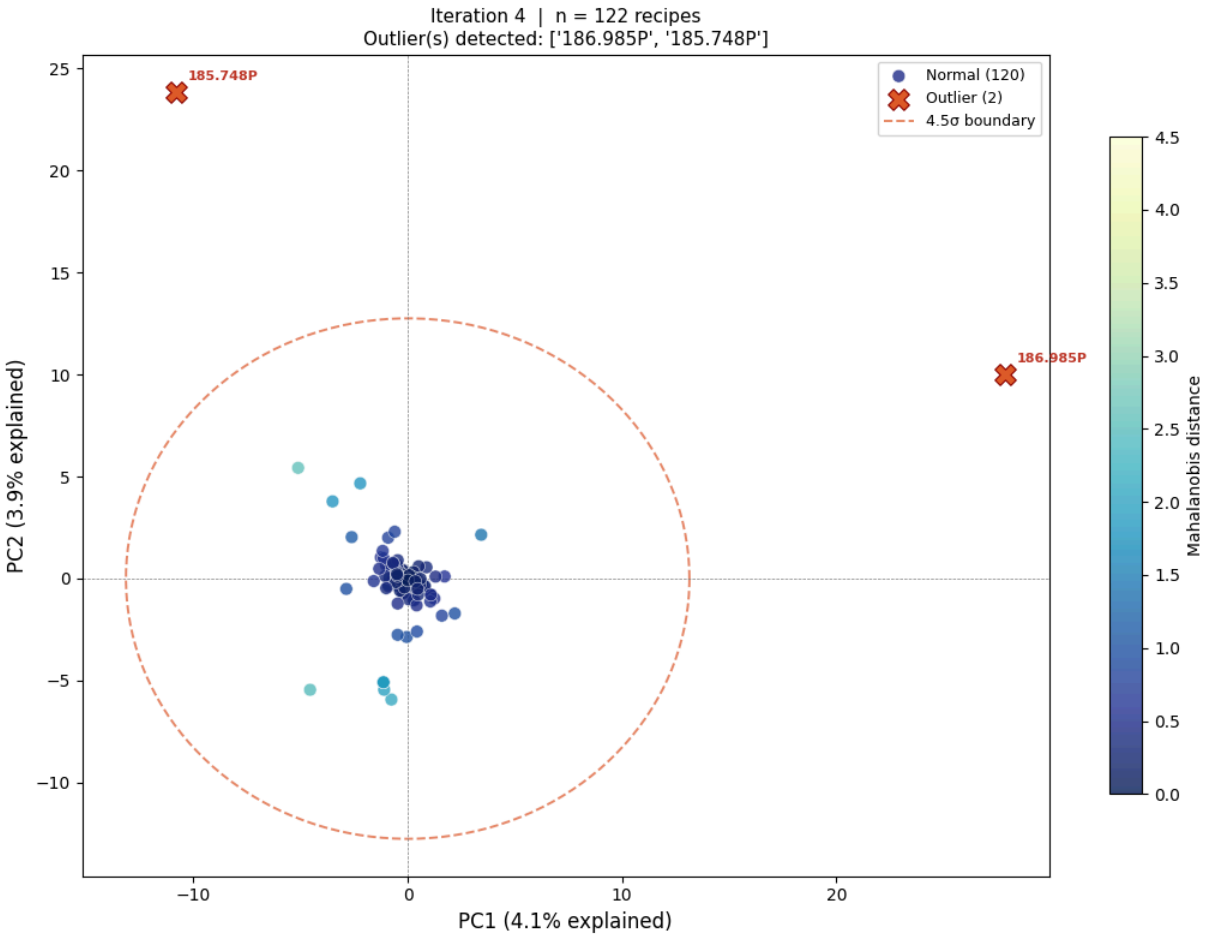
Saved: pca_v2_noOT_iter01.png
Iteration 1: removing 2 outlier(s): ['187.004P', '185.309P']



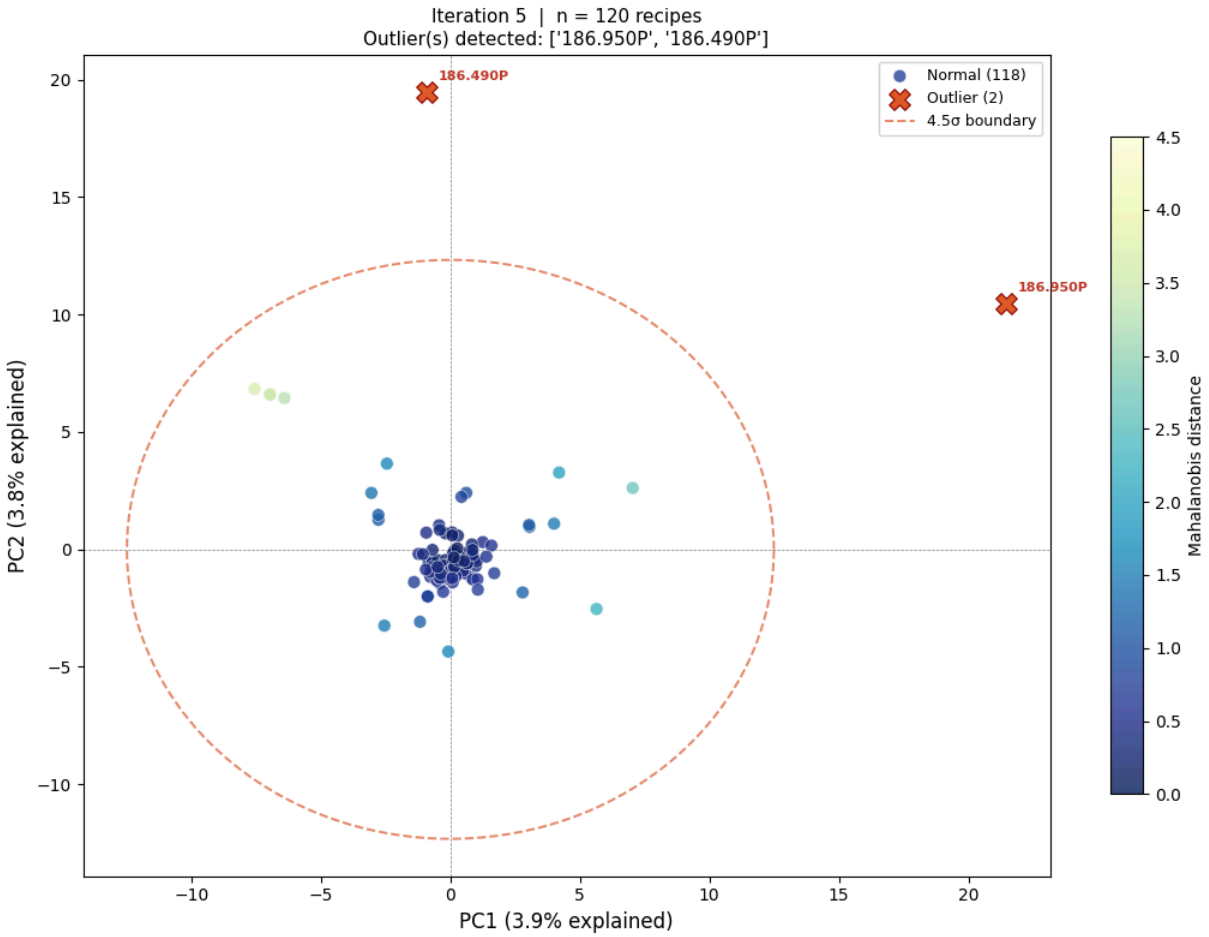
Saved: pca_v2_noOT_iter02.png
Iteration 2: removing 3 outlier(s): ['187.015P', '187.482P', '185.382P']



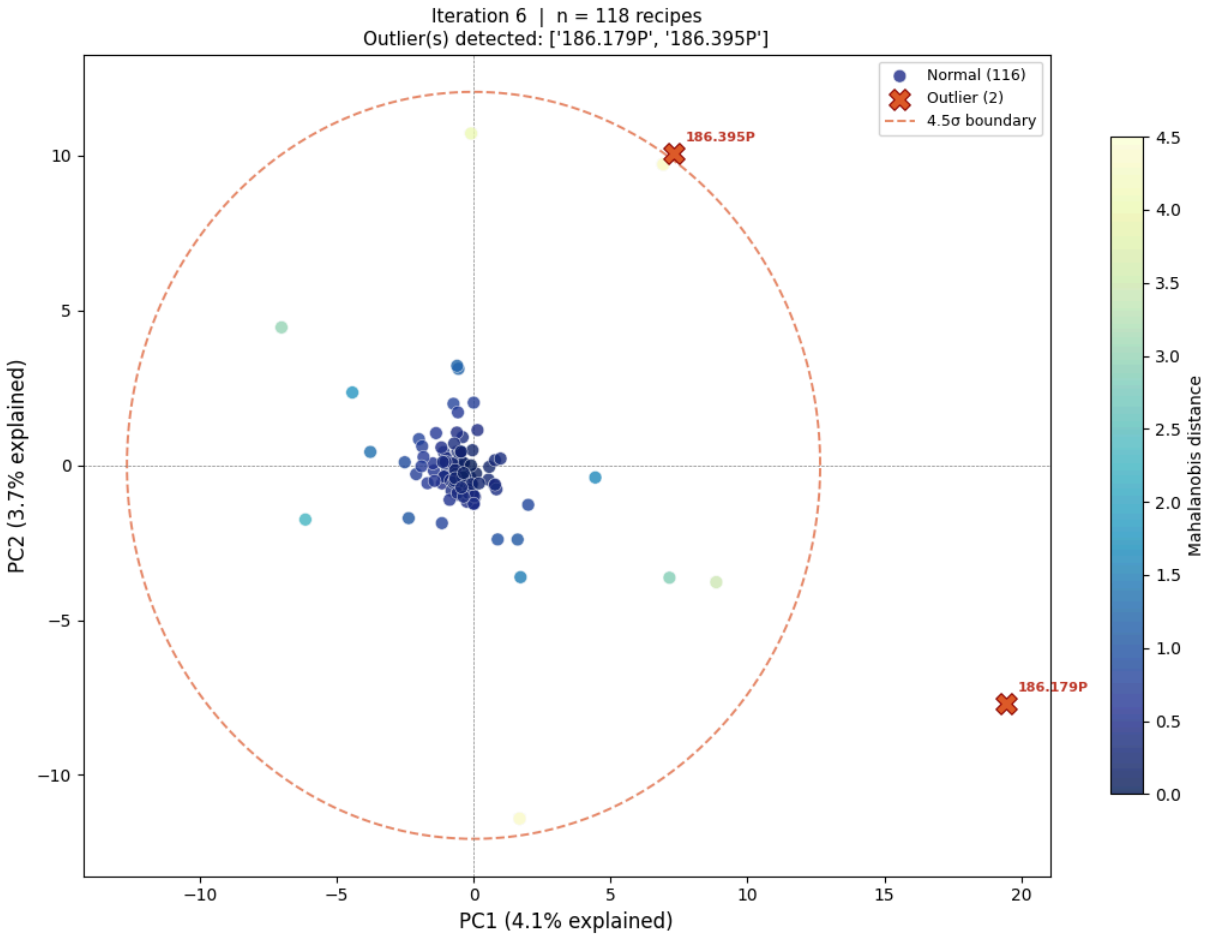
Saved: pca_v2_noOT_iter03.png
Iteration 3: removing 2 outlier(s): ['188.462KHP', '185.028']



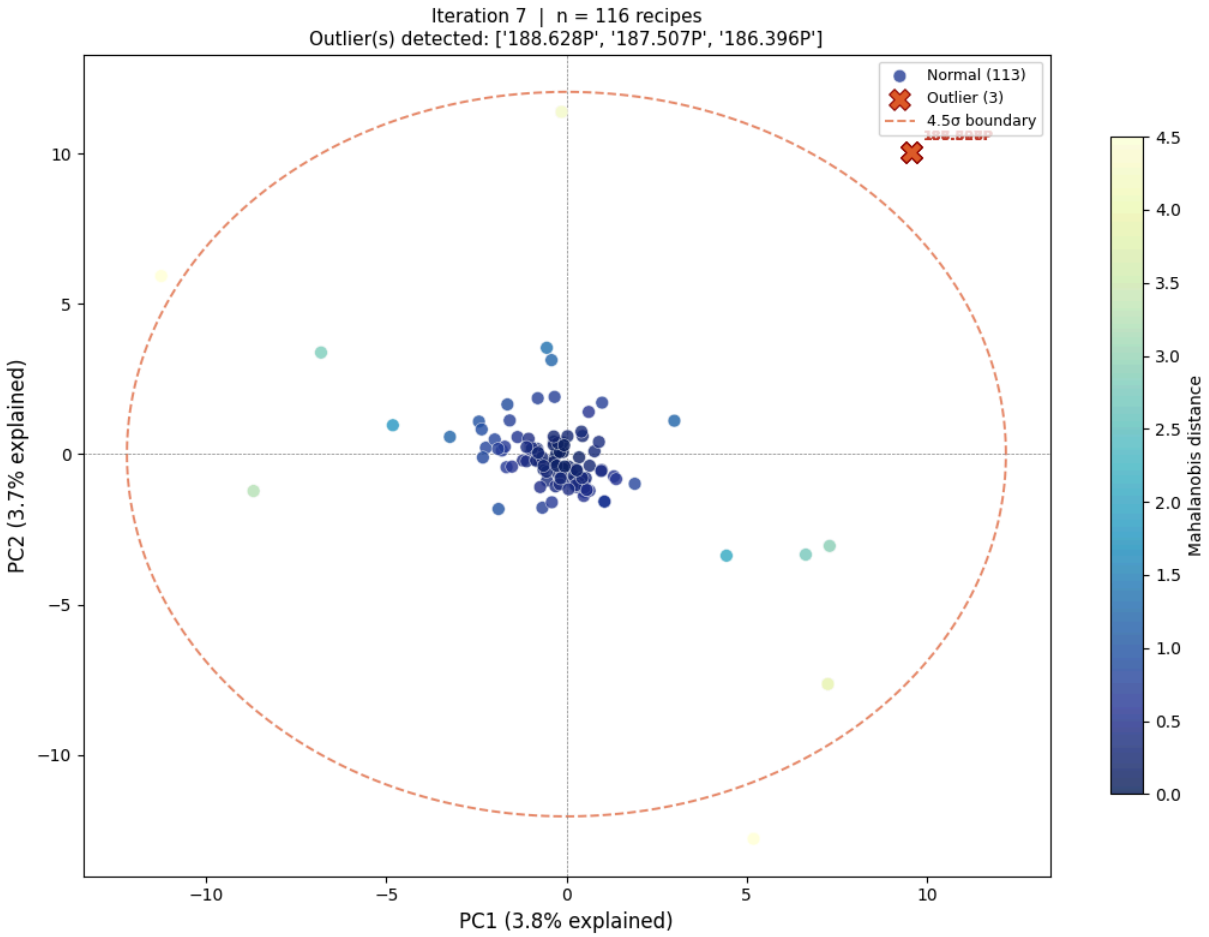
Saved: pca_v2_noOT_iter04.png
Iteration 4: removing 2 outlier(s): ['186.985P', '185.748P']



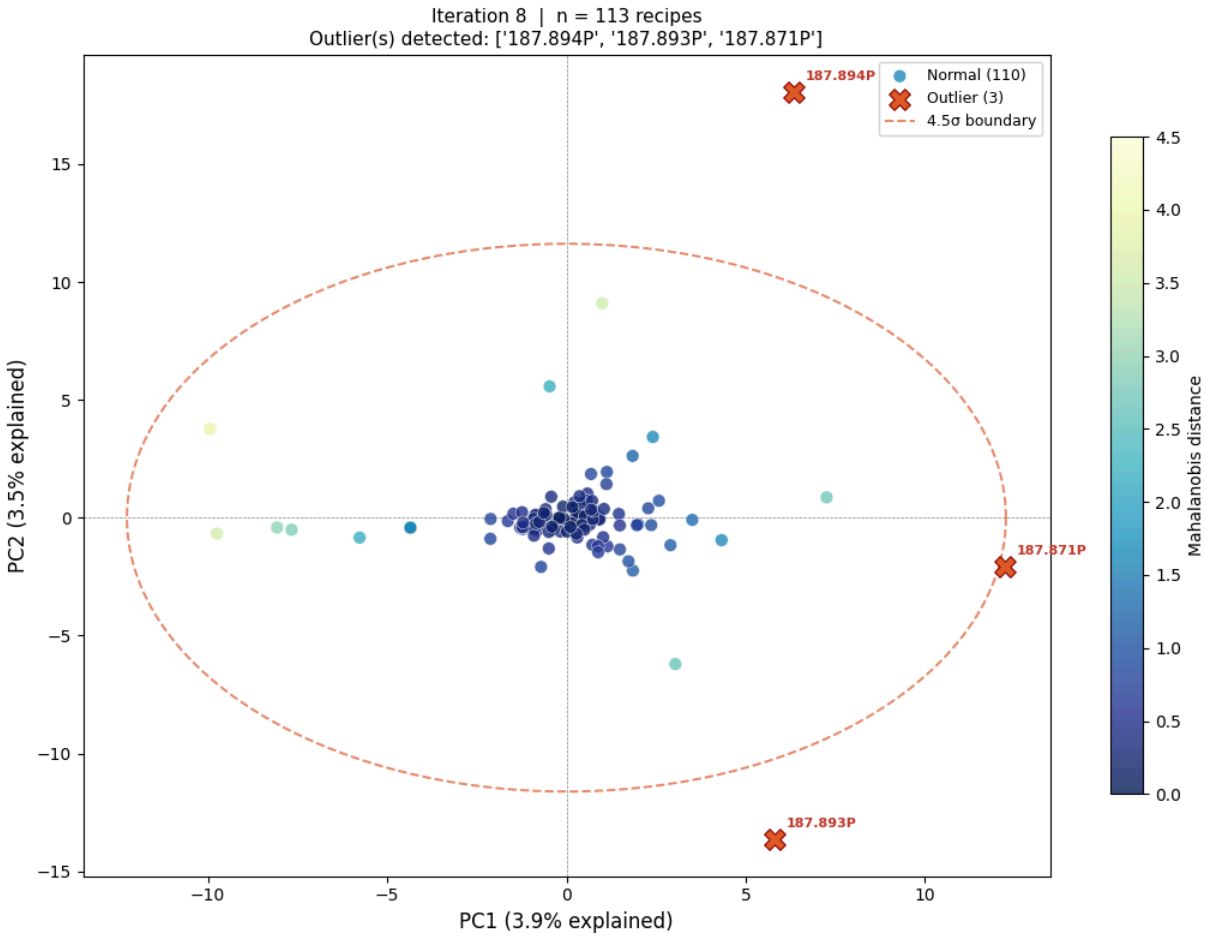
Saved: pca_v2_noOT_iter05.png
Iteration 5: removing 2 outlier(s): ['186.950P', '186.490P']



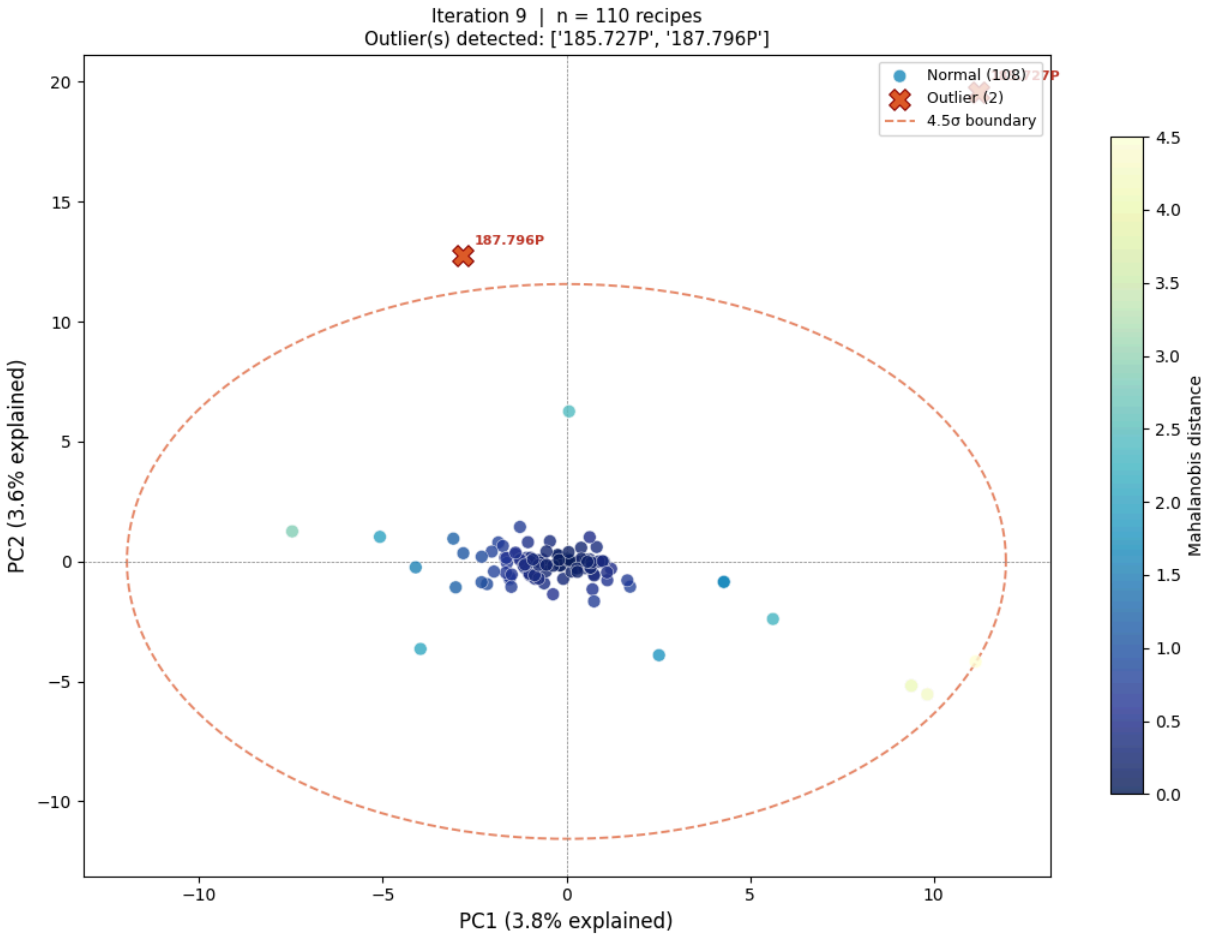
Saved: pca_v2_noOT_iter06.png
Iteration 6: removing 2 outlier(s): ['186.179P', '186.395P']



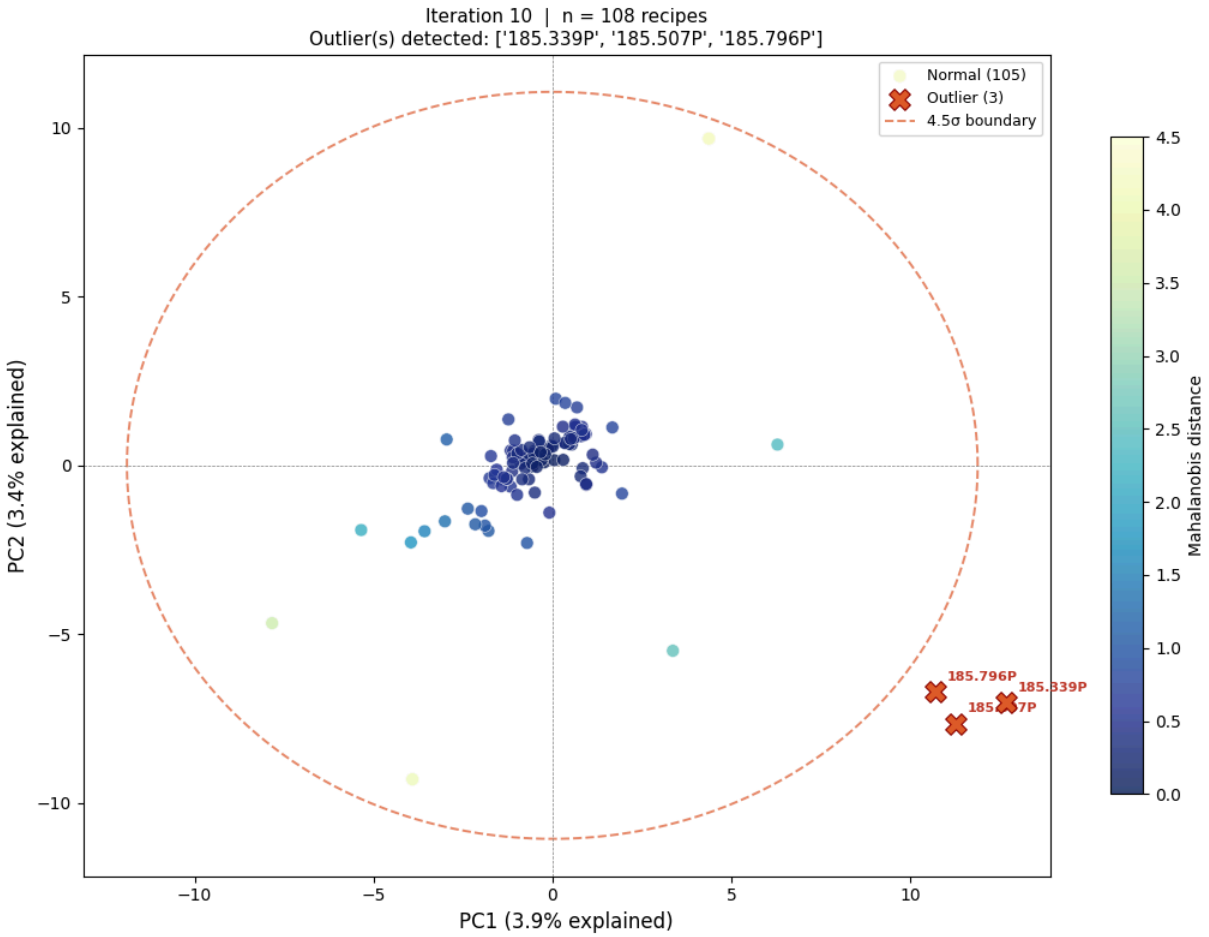
Saved: pca_v2_noOT_iter07.png
Iteration 7: removing 3 outlier(s): ['188.628P', '187.507P', '186.396P']



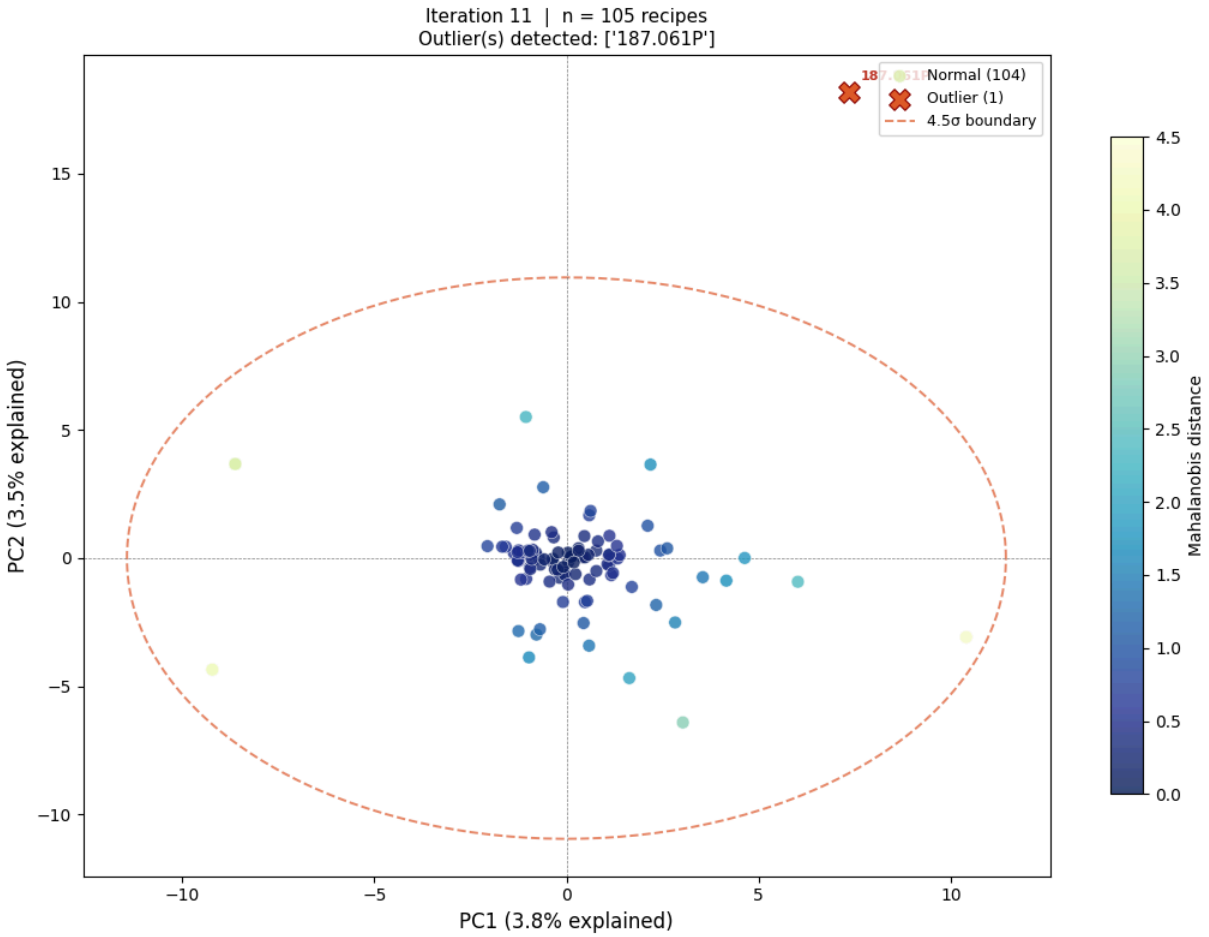
Saved: pca_v2_noOT_iter08.png
Iteration 8: removing 3 outlier(s): ['187.894P', '187.893P', '187.871P']



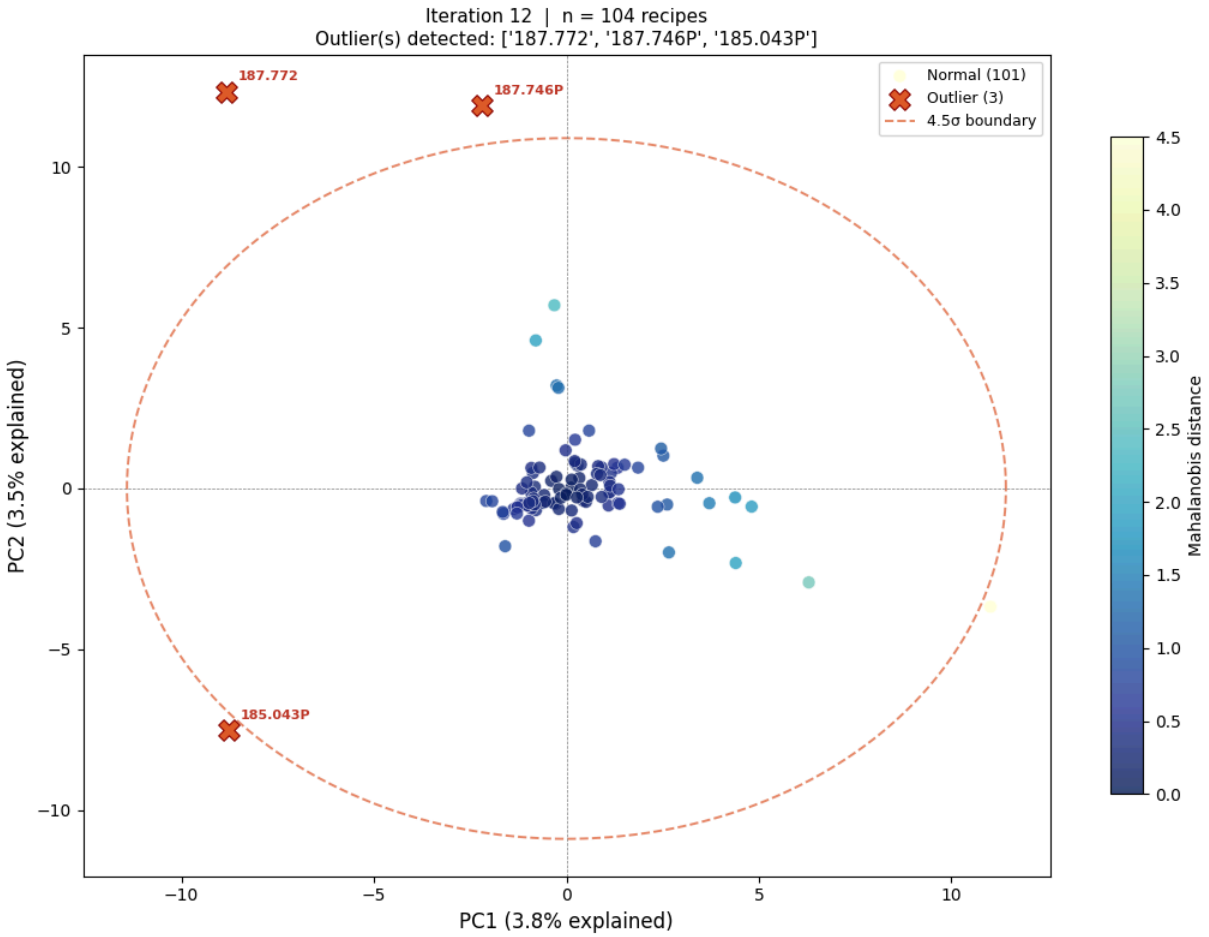
Saved: pca_v2_noOT_iter09.png
Iteration 9: removing 2 outlier(s): ['185.727P', '187.796P']



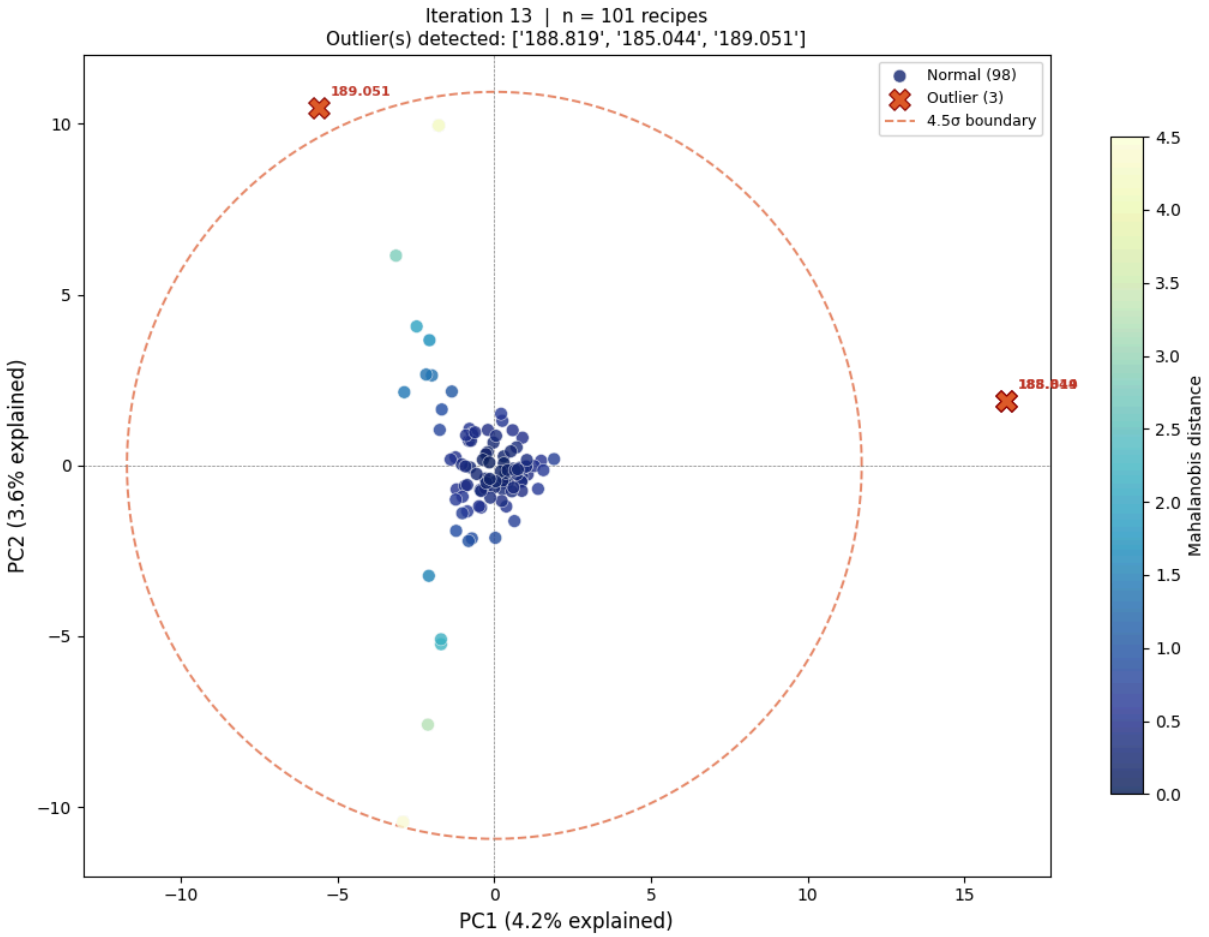

```
Saved: pca_v2_noOT_iter10.png
Iteration 10: removing 3 outlier(s): ['185.339P', '185.507P', '185.796P']
```



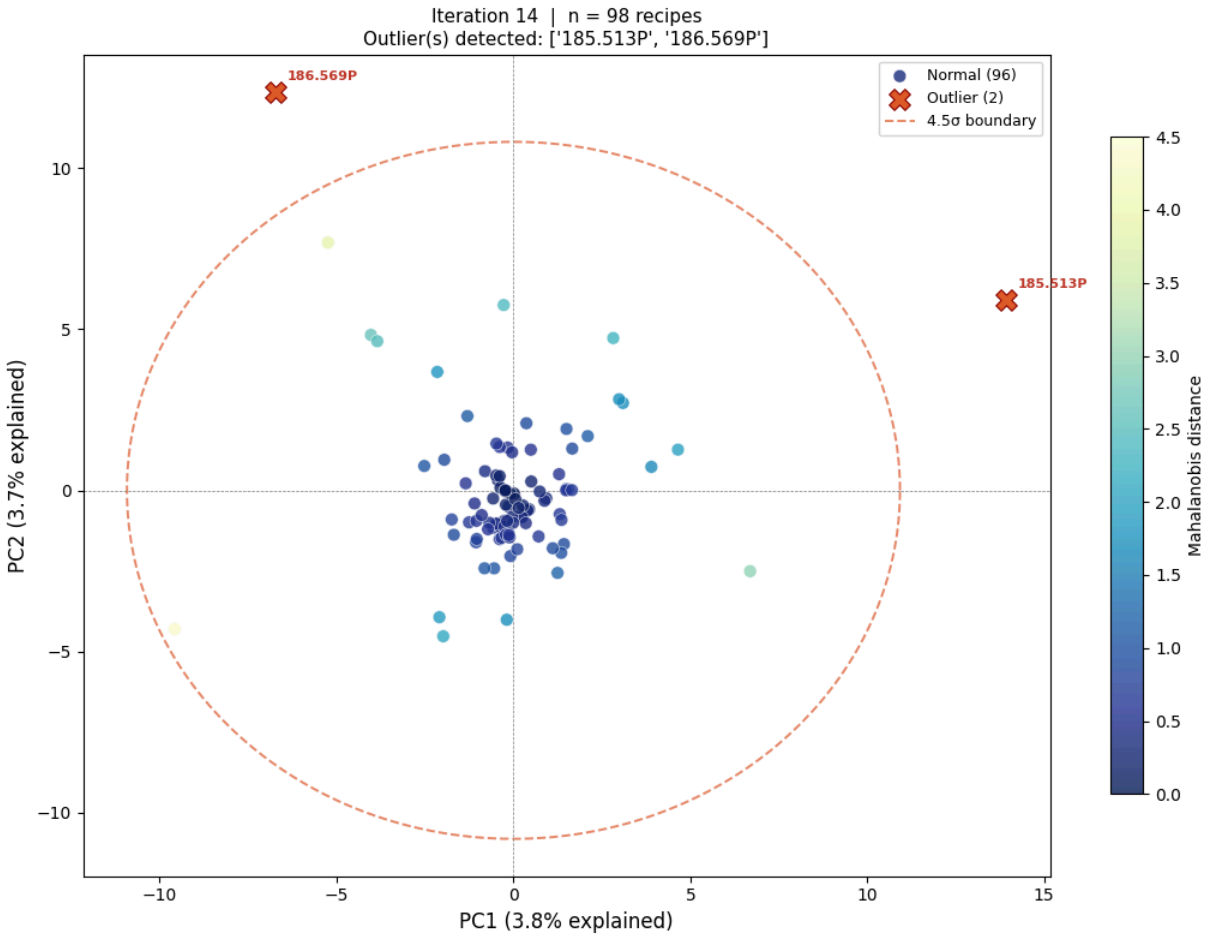

```
Saved: pca_v2_noOT_iter11.png
Iteration 11: removing 1 outlier(s): ['187.061P']
```



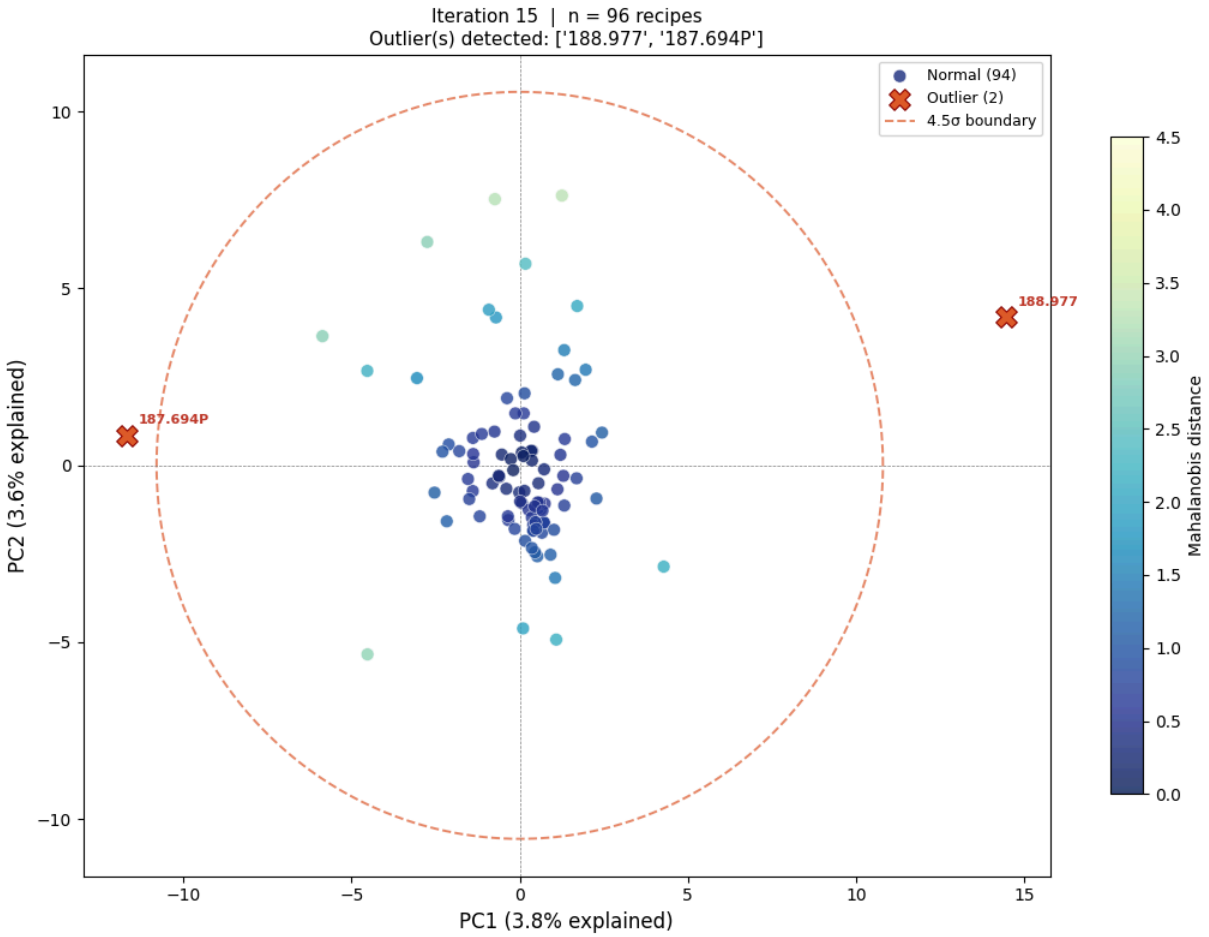

```
Saved: pca_v2_noOT_iter12.png
Iteration 12: removing 3 outlier(s): ['187.772', '187.746P', '185.043P']
```



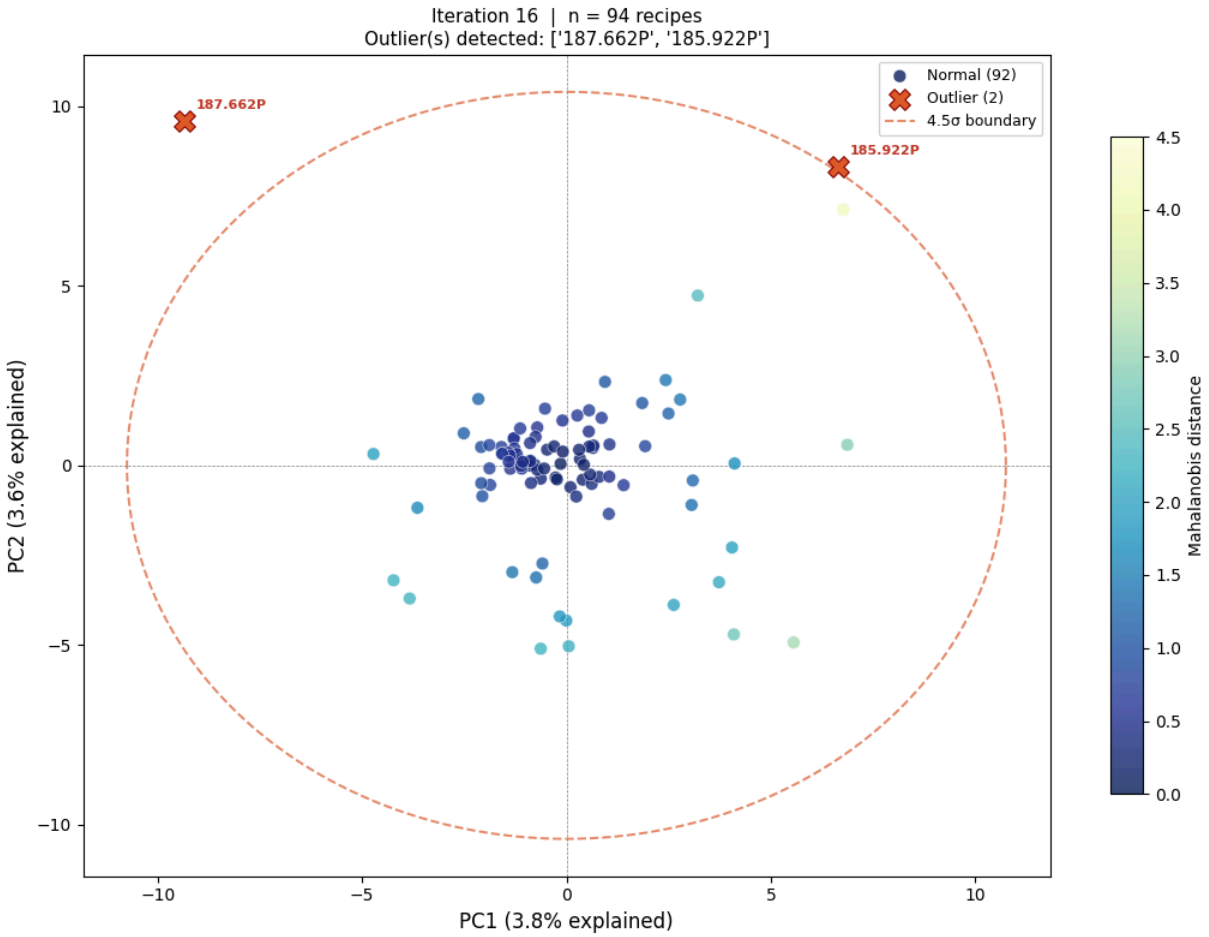
Saved: pca_v2_noOT_iter13.png
Iteration 13: removing 3 outlier(s): ['188.819', '185.044', '189.051']



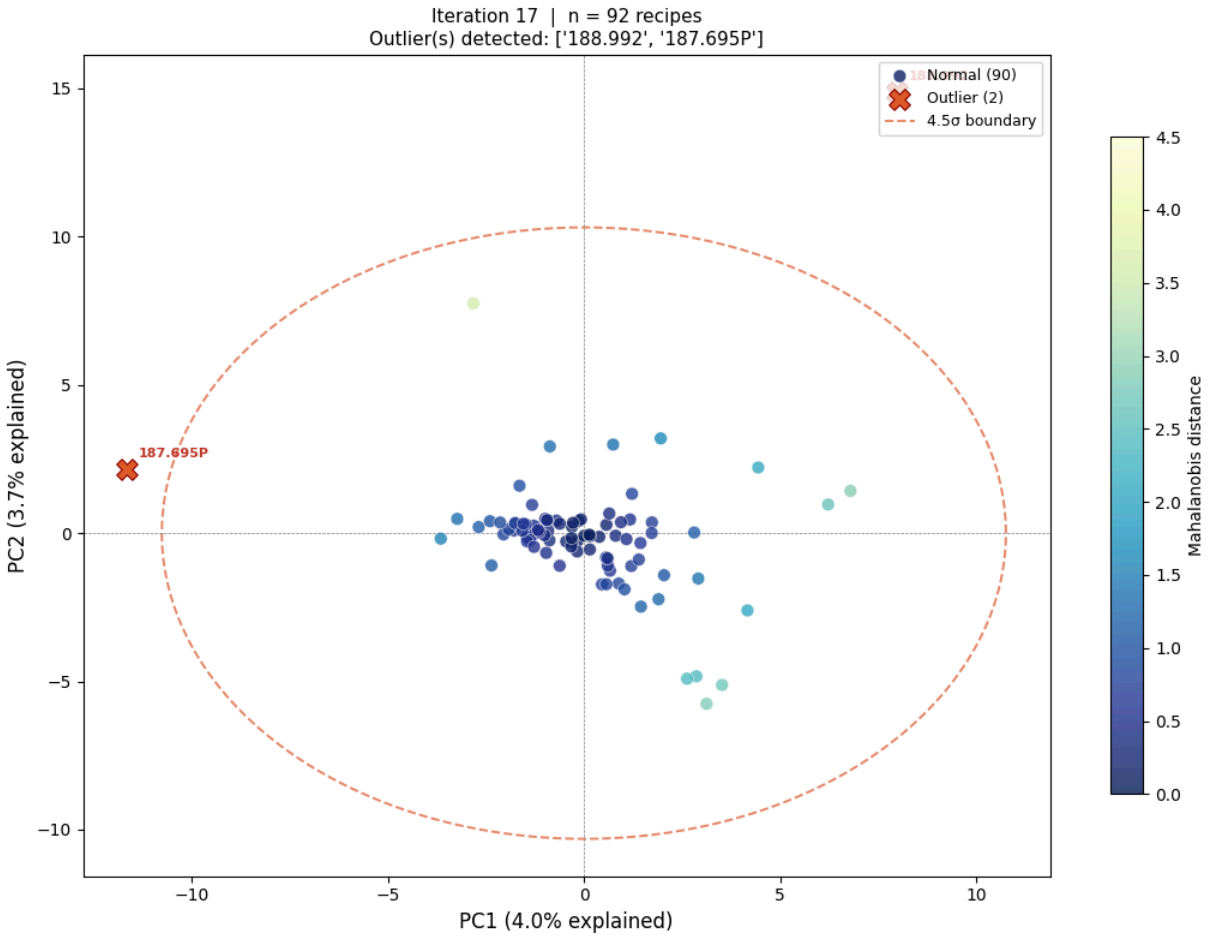
Saved: pca_v2_noOT_iter14.png
Iteration 14: removing 2 outlier(s): ['185.513P', '186.569P']



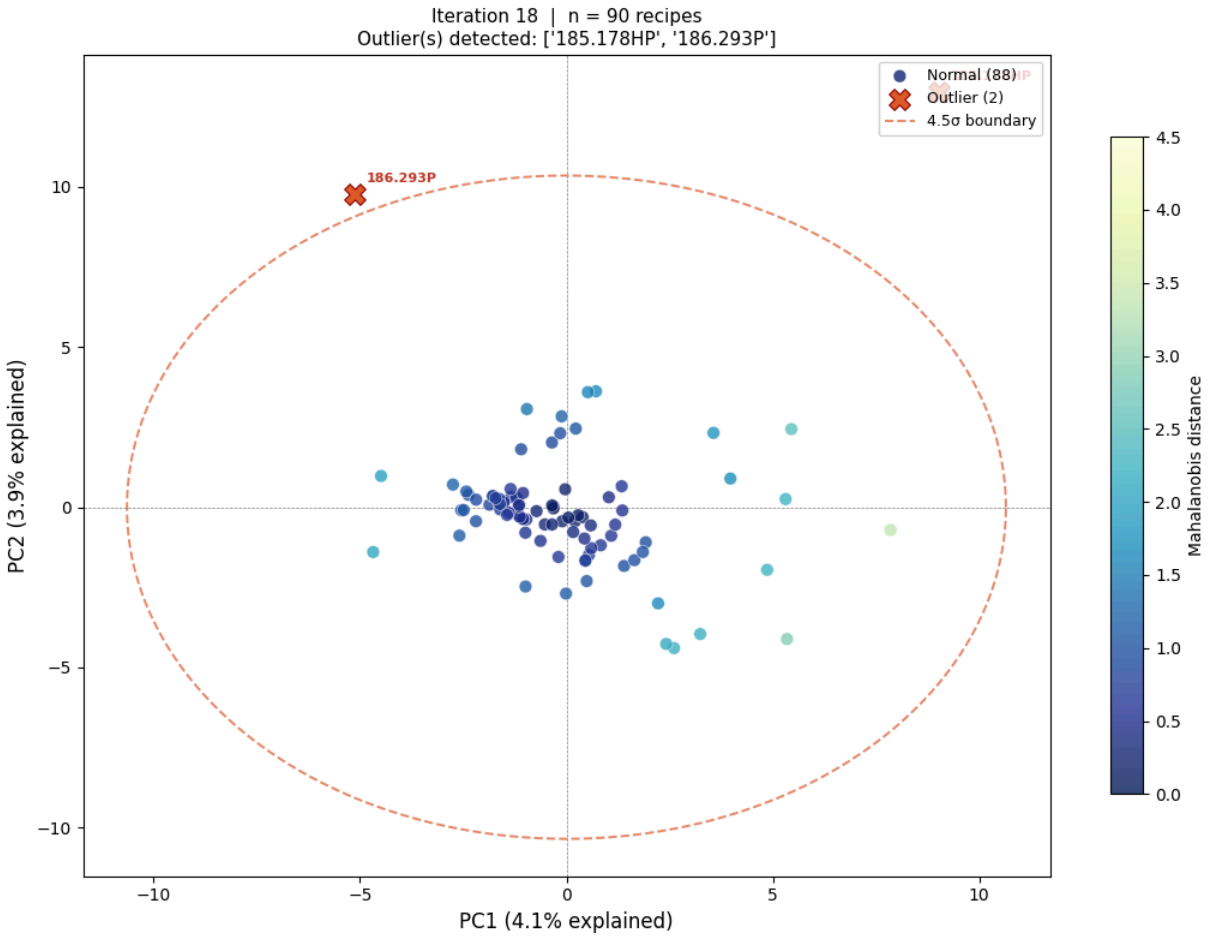

```
Saved: pca_v2_noOT_iter15.png
Iteration 15: removing 2 outlier(s): ['188.977', '187.694P']
```



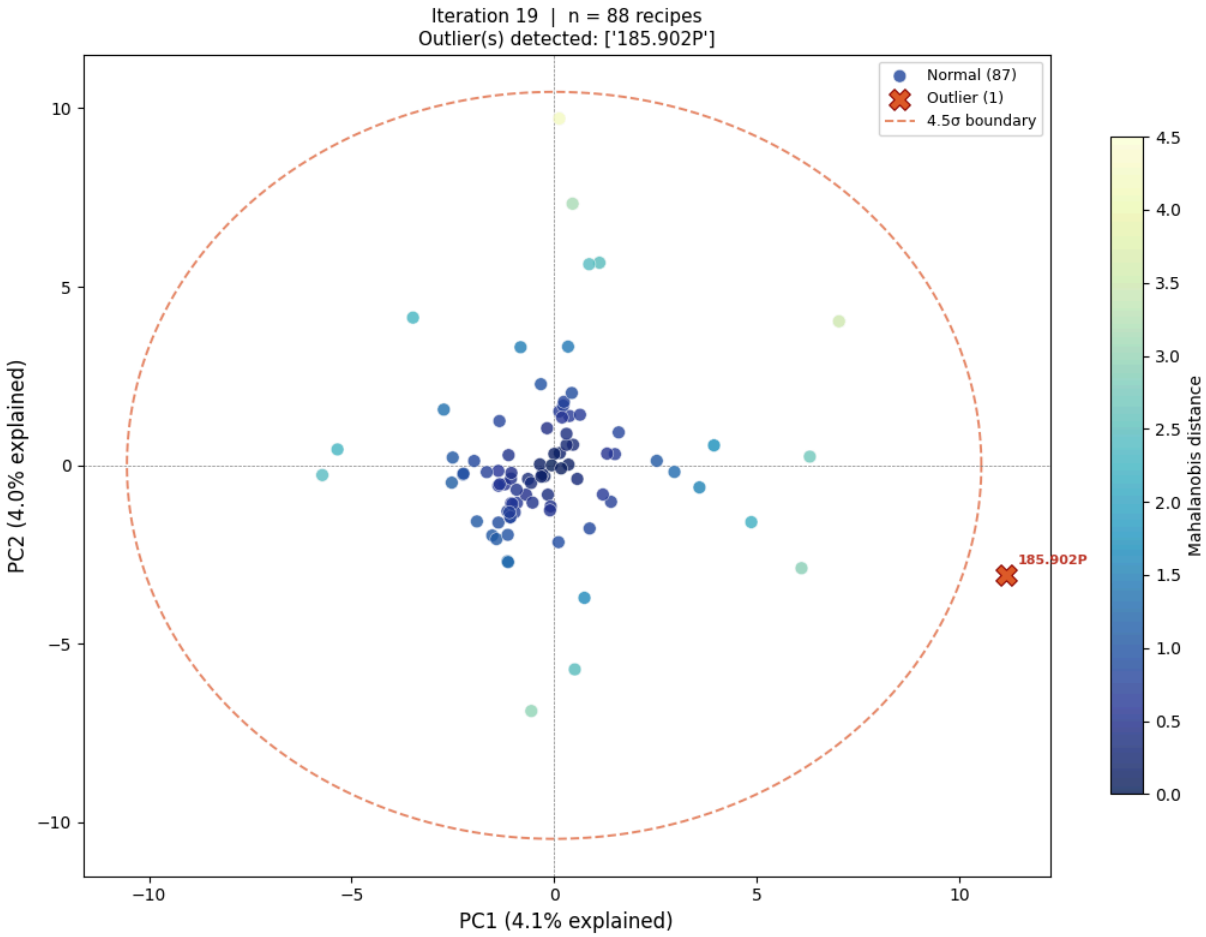

```
Saved: pca_v2_noOT_iter16.png
Iteration 16: removing 2 outlier(s): ['187.662P', '185.922P']
```



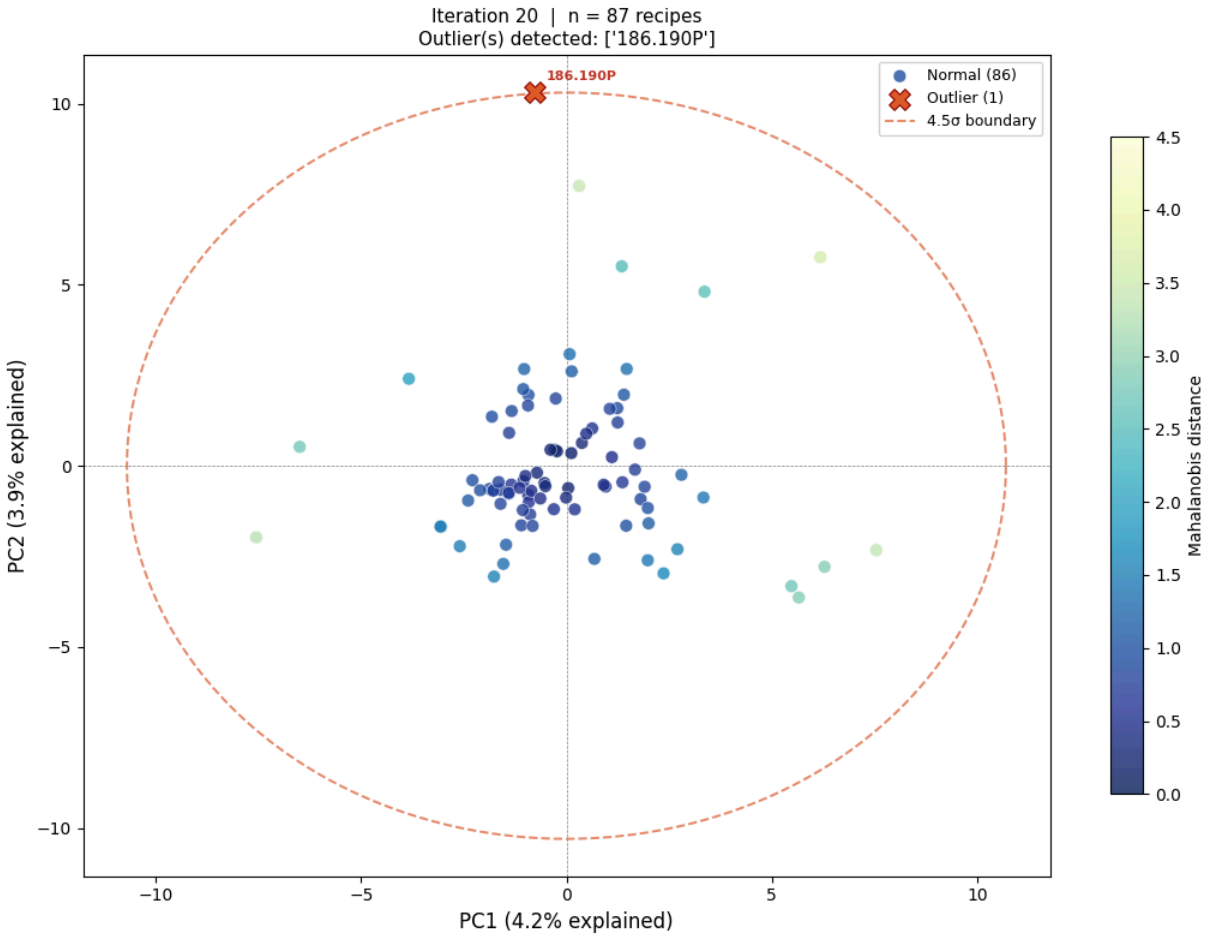
Saved: pca_v2_noOT_iter17.png
Iteration 17: removing 2 outlier(s): ['188.992', '187.695P']



Saved: pca_v2_noOT_iter18.png
Iteration 18: removing 2 outlier(s): ['185.178HP', '186.293P']




```
Saved: pca_v2_noOT_iter19.png
Iteration 19: removing 1 outlier(s): ['185.902P']
```




```
Saved: pca_v2_noOT_iter20.png
Iteration 20: removing 1 outlier(s): ['186.190P']

⚠ Maximum 20 iterations reached — stopping.
  Remaining dataset: 86 recipes

Total removed: 43 recipe(s)
Removed list: [(1, '187.004P'), (1, '185.309P'), (2, '187.015P'), (2, '187.4
82P'), (2, '185.382P'), (3, '188.462KHP'), (3, '185.028'), (4, '186.985P'),
(4, '185.748P'), (5, '186.950P'), (5, '186.490P'), (6, '186.179P'), (6, '18
6.395P'), (7, '188.628P'), (7, '187.507P'), (7, '186.396P'), (8, '187.894
P'), (8, '187.893P'), (8, '187.871P'), (9, '185.727P'), (9, '187.796P'), (1
0, '185.339P'), (10, '185.507P'), (10, '185.796P'), (11, '187.061P'), (12,
'187.772'), (12, '187.746P'), (12, '185.043P'), (13, '188.819'), (13, '185.0
44'), (13, '189.051'), (14, '185.513P'), (14, '186.569P'), (15, '188.977'),
(15, '187.694P'), (16, '187.662P'), (16, '185.922P'), (17, '188.992'), (17,
'187.695P'), (18, '185.178HP'), (18, '186.293P'), (19, '185.902P'), (20, '18
6.190P')]
```

In [8]:

```python
# ── Iteration summary table ──────────────────────────────────────────
iter_df = pd.DataFrame(iteration_log)
print('=== Iteration Log ===')
print(iter_df.to_string(index=False))

# Set up convenient aliases used by all subsequent sections
scaler_oav   = final_scaler
pca_oav      = final_pca
scores_oav   = final_scores
```

```
ev           = final_ev
recipes_oav  = final_active_recipes
X_oav_scaled = final_X_sc
loadings     = pca_oav.components_
cas_labels   = list(pivot_oav.columns)
cas_name     = df.groupby('CAS-Nr.')['Name'].first().to_dict()

# Final pivot (only clean recipes)
final_pivot  = pivot_oav.loc[recipes_oav]

print(f'\nFinal dataset: {len(recipes_oav)} recipes × {final_pivot.shape[1]}
print(f'Explained variance: PC1={ev[0]:.1f}%  PC2={ev[1]:.1f}%  PC3={ev[2]:.
```

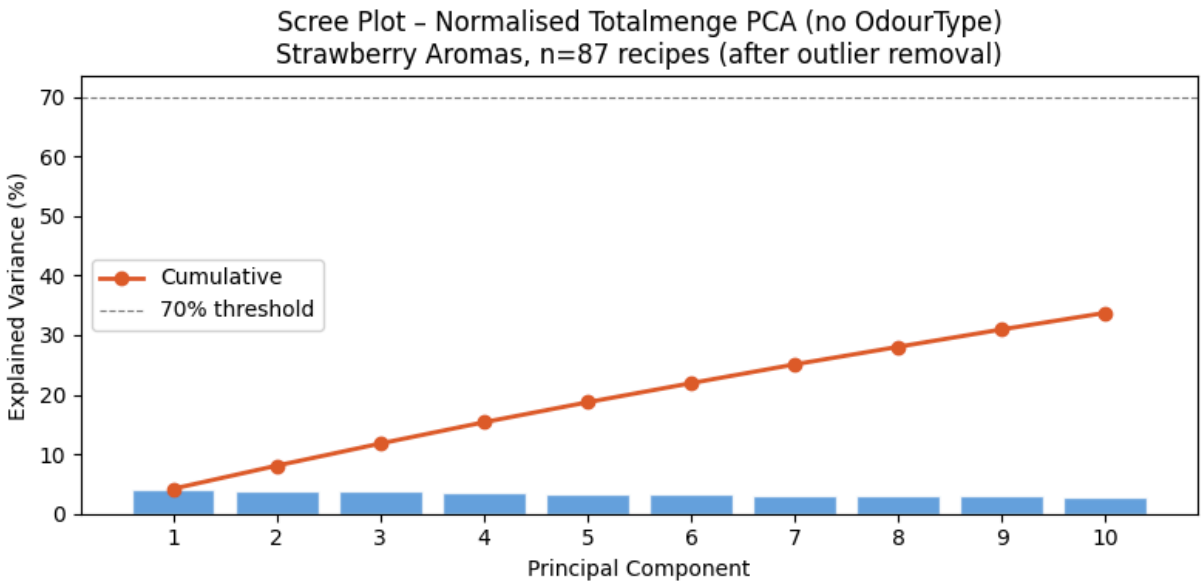
```
=== Iteration Log ===
 iteration  n_recipes                      outliers  n_outliers  ev_PC1  ev
_PC2  ev_cum3
         1        129           [187.004P, 185.309P]           2     5.2
4.8     14.0
         2        127 [187.015P, 187.482P, 185.382P]           3     4.3
4.1     12.0
         3        124          [188.462KHP, 185.028]           2     4.0
3.7     11.3
         4        122           [186.985P, 185.748P]           2     4.1
3.9     11.7
         5        120           [186.950P, 186.490P]           2     3.9
3.8     11.2
         6        118           [186.179P, 186.395P]           2     4.1
3.7     11.3
         7        116 [188.628P, 187.507P, 186.396P]           3     3.8
3.7     10.9
         8        113 [187.894P, 187.893P, 187.871P]           3     3.9
3.5     10.8
         9        110           [185.727P, 187.796P]           2     3.8
3.6     10.7
        10        108 [185.339P, 185.507P, 185.796P]           3     3.9
3.4     10.6
        11        105                     [187.061P]           1     3.8
3.5     10.7
        12        104  [187.772, 187.746P, 185.043P]           3     3.8
3.5     10.7
        13        101    [188.819, 185.044, 189.051]           3     4.2
3.6     11.3
        14         98           [185.513P, 186.569P]           2     3.8
3.7     11.2
        15         96            [188.977, 187.694P]           2     3.8
3.6     11.0
        16         94           [187.662P, 185.922P]           2     3.8
3.6     11.0
        17         92            [188.992, 187.695P]           2     4.0
3.7     11.3
        18         90          [185.178HP, 186.293P]           2     4.1
3.9     11.7
        19         88                     [185.902P]           1     4.1
4.0     11.8
        20         87                     [186.190P]           1     4.2
3.9     11.8

Final dataset: 87 recipes × 225 CAS
Explained variance: PC1=4.2%  PC2=3.9%  PC3=3.7%
```

3 · Final PCA Analysis (Clean Dataset)

In [9]:

```
# ── Scree plot ──────────────────────────────────────────────────────
fig, ax = plt.subplots(figsize=(8, 4))
ax.bar(range(1, len(ev)+1), ev, color='#4A90D9', alpha=0.85, edgecolor='whit
ax.plot(range(1, len(ev)+1), np.cumsum(ev), 'o-', color='#E05A2B', lw=2, lab
ax.axhline(70, ls='--', color='grey', lw=0.8, label='70% threshold')
ax.set_xlabel('Principal Component')
```

```
ax.set_ylabel('Explained Variance (%)')
ax.set_title(
    f'Scree Plot — Normalised Totalmenge PCA (no OdourType)\n'
    f'Strawberry Aromas, n={len(recipes_oav)} recipes (after outlier removal
)
ax.set_xticks(range(1, len(ev)+1))
ax.legend()
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_scree.png', dpi=150)
plt.show()
print('Saved: pca_v2_noOT_scree.png')
```

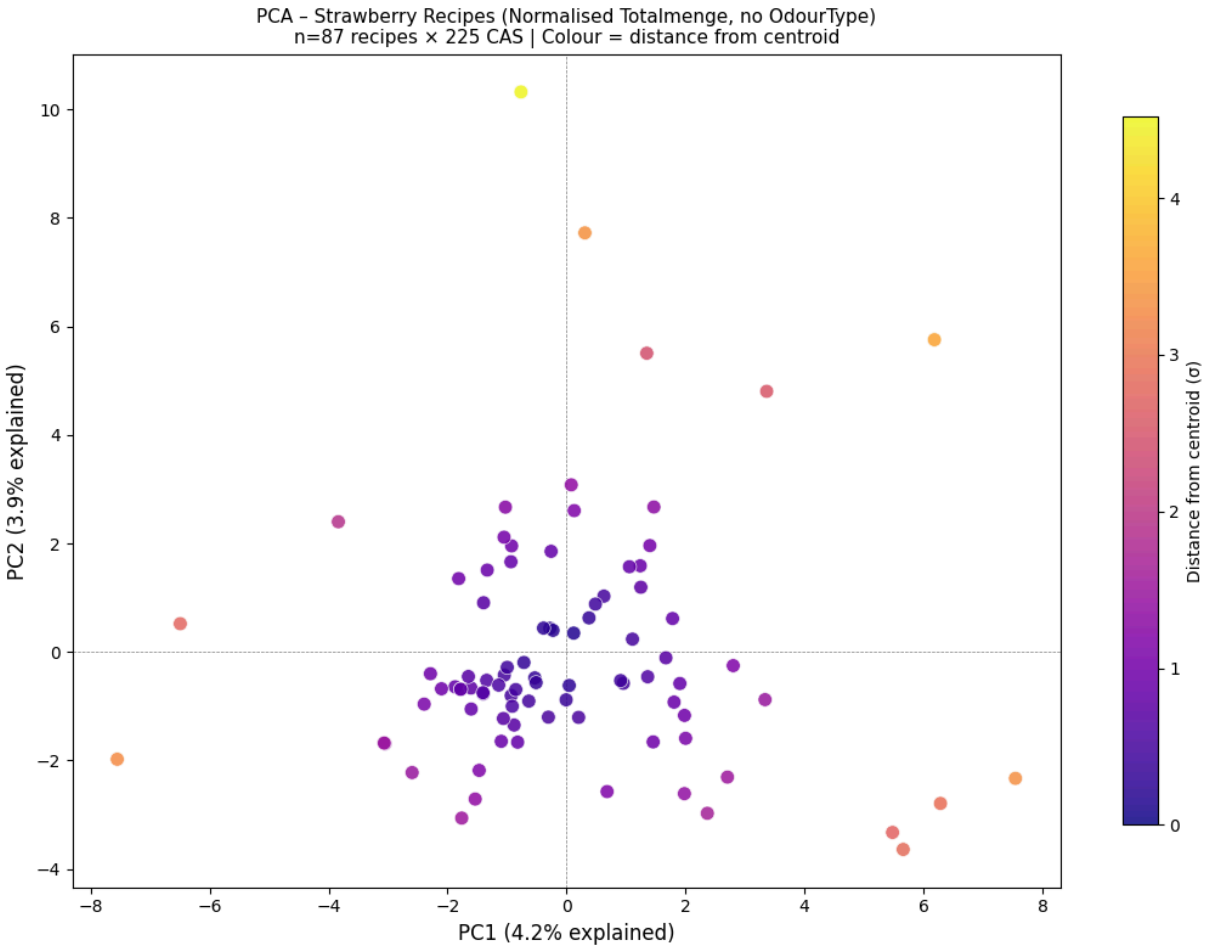



```
Saved: pca_v2_noOT_scree.png
```

In [10]:

```
# ── PC1 vs PC2 scatter — coloured by Mahalanobis distance from centroid ──
s1_f  = scores_oav[:, 0]
s2_f  = scores_oav[:, 1]
z1_f  = (s1_f - s1_f.mean()) / (s1_f.std() + 1e-12)
z2_f  = (s2_f - s2_f.mean()) / (s2_f.std() + 1e-12)
dist_f = np.sqrt(z1_f**2 + z2_f**2)

fig, ax = plt.subplots(figsize=(11, 8))
sc = ax.scatter(
    s1_f, s2_f,
    c=dist_f, cmap='plasma', vmin=0, vmax=dist_f.max(),
    s=70, alpha=0.85, edgecolors='white', linewidths=0.5, zorder=3
)
plt.colorbar(sc, ax=ax, label='Distance from centroid (σ)', shrink=0.85)

ax.axhline(0, color='grey', lw=0.5, ls='--')
ax.axvline(0, color='grey', lw=0.5, ls='--')
ax.set_xlabel(f'PC1 ({ev[0]:.1f}% explained)', fontsize=12)
ax.set_ylabel(f'PC2 ({ev[1]:.1f}% explained)', fontsize=12)
ax.set_title(
    f'PCA — Strawberry Recipes (Normalised Totalmenge, no OdourType)\n'
    f'n={len(recipes_oav)} recipes × {final_pivot.shape[1]} CAS | '
    f'Colour = distance from centroid',
    fontsize=11
```

```
)
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_pc1pc2.png', dpi=150, bbox_inches='tight'
plt.show()
print('Saved: pca_v2_noOT_pc1pc2.png')
```




```
Saved: pca_v2_noOT_pc1pc2.png
```

4 · Biplot – Top Ingredient Loadings

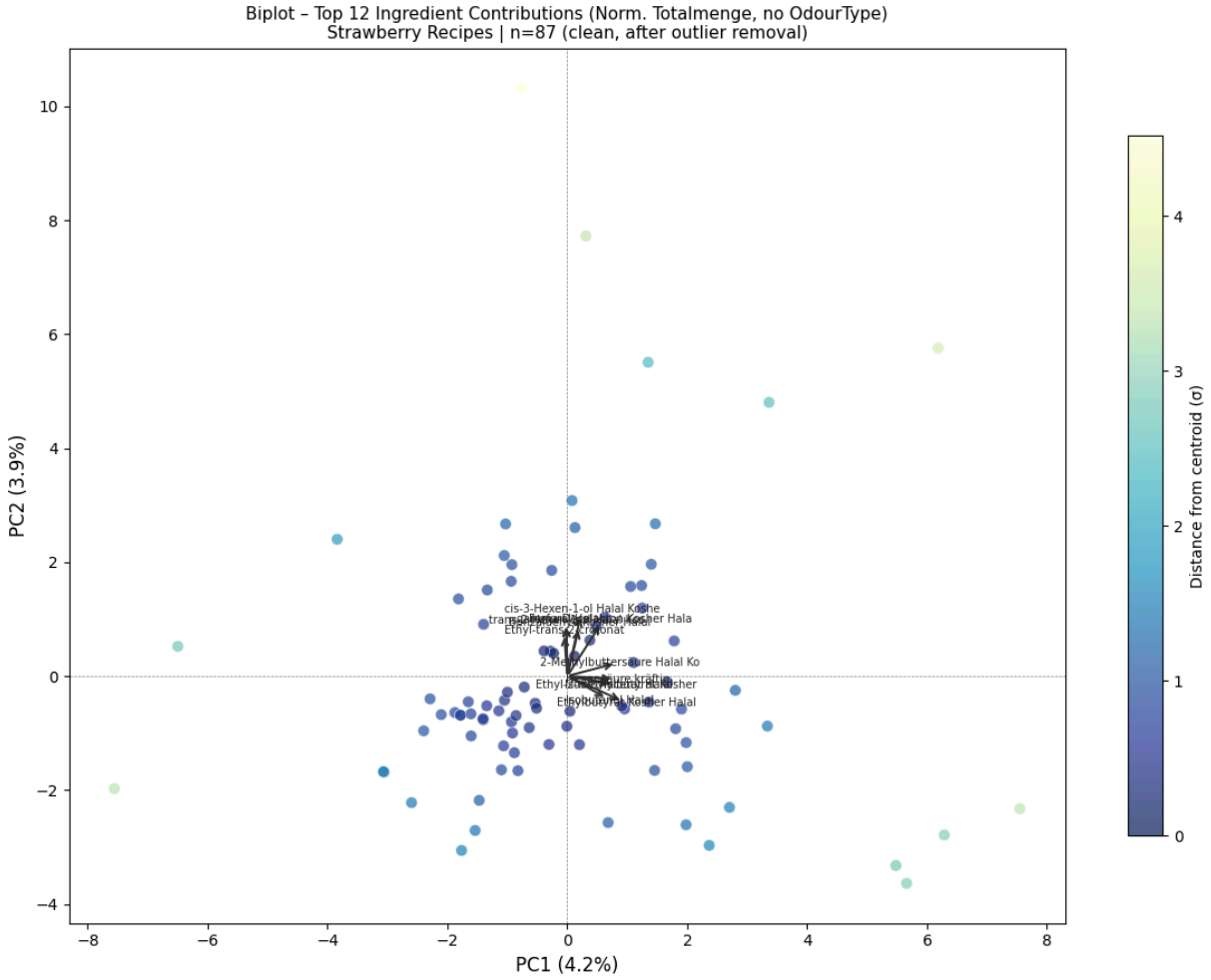
In [11]:
```
N_ARROWS  = 12
importance = np.sqrt(loadings[0]**2 + loadings[1]**2)
top_idx    = np.argsort(importance)[::-1][:N_ARROWS]

print(f'Top {N_ARROWS} ingredients driving PC1+PC2:')
for i in top_idx:
    cas  = cas_labels[i]
    name = cas_name.get(cas, cas)
    print(f'  {cas:15s} {name[:45]:45s}  L1={loadings[0,i]:+.3f}  L2={loadir
```

```
Top 12 ingredients driving PC1+PC2:
  928-96-1        cis-3-Hexen-1-ol Halal Kosher                 L1=+0.065
L2=+0.310
  706-14-9        gamma-Decalacton Kosher Halal                 L1=+0.163
L2=+0.266
  105-54-4        Ethylbutyrat Kosher Halal                     L1=+0.263
L2=-0.126
  98-01-1         Furfural Halal                                L1=-0.006
L2=+0.263
  2305-21-7       trans-2-Hexen-1-ol Halal Kosher               L1=-0.004
L2=+0.257
  100-52-7        Benzaldehyd Kosher Halal                      L1=+0.057
L2=+0.246
  116-53-0        2-Methylbuttersäure Halal Kosher              L1=+0.233
L2=+0.064
  123-92-2        Isoamylacetat Halal                           L1=+0.225
L2=-0.041
  142-62-1        Hexansäure kräftig                            L1=+0.224
L2=-0.015
  7452-79-1       Ethyl-2-methylbutyrat Kosher Halal            L1=+0.216
L2=-0.043
  78-83-1         Isobutanol Halal                              L1=+0.189
L2=-0.112
  10544-63-5      Ethyl-trans-2-crotonat                        L1=-0.015
L2=+0.212
```

In [12]:

```python
# ── Biplot ──────────────────────────────────────────────────────────
SCALE = 3.5

fig, ax = plt.subplots(figsize=(12, 9))

# Scatter recipes (coloured by Mahalanobis distance)
sc = ax.scatter(
    scores_oav[:, 0], scores_oav[:, 1],
    c=dist_f, cmap='YlGnBu_r', vmin=0, vmax=dist_f.max(),
    s=55, alpha=0.70, edgecolors='white', linewidths=0.4, zorder=3
)
plt.colorbar(sc, ax=ax, label='Distance from centroid (σ)', shrink=0.8)

# Arrow overlays for top ingredients
for i in top_idx:
    cas   = cas_labels[i]
    name  = cas_name.get(cas, cas)
    short = name.split(',')[0][:28]
    lx, ly = loadings[0, i] * SCALE, loadings[1, i] * SCALE
    ax.annotate('', xy=(lx, ly), xytext=(0, 0),
                arrowprops=dict(arrowstyle='->', color='#333333', lw=1.4))
    ax.text(lx * 1.08, ly * 1.08, short, fontsize=7, color='#222222',
            ha='center', va='center')

ax.axhline(0, color='grey', lw=0.5, ls='--')
ax.axvline(0, color='grey', lw=0.5, ls='--')
ax.set_xlabel(f'PC1 ({ev[0]:.1f}%)', fontsize=12)
ax.set_ylabel(f'PC2 ({ev[1]:.1f}%)', fontsize=12)
ax.set_title(
    f'Biplot — Top {N_ARROWS} Ingredient Contributions (Norm. Totalmenge, nc
```

```
    f'Strawberry Recipes | n={len(recipes_oav)} (clean, after outlier removal
    fontsize=11
)
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_biplot.png', dpi=150, bbox_inches='tight'
plt.show()
print('Saved: pca_v2_noOT_biplot.png')
```

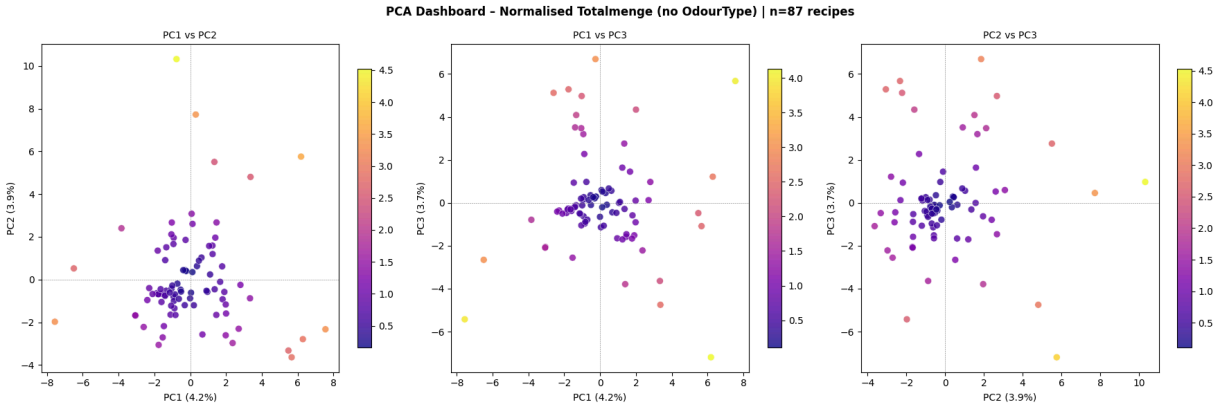


Saved: pca_v2_noOT_biplot.png

5 · PC1–PC3 Dashboard (3-Panel)

In [13]:
```
# ── PC1 vs PC2, PC1 vs PC3, PC2 vs PC3 ──────────────────────────────────
fig, axes = plt.subplots(1, 3, figsize=(18, 6))
pairs = [(0, 1), (0, 2), (1, 2)]

for ax, (i, j) in zip(axes, pairs):
    si = scores_oav[:, i]
    sj = scores_oav[:, j]
    zi = (si - si.mean()) / (si.std() + 1e-12)
    zj = (sj - sj.mean()) / (sj.std() + 1e-12)
    d_ij = np.sqrt(zi**2 + zj**2)
    sc = ax.scatter(
        si, sj,
        c=d_ij, cmap='plasma',
        s=50, alpha=0.80, edgecolors='white', linewidths=0.4
```

```
    )
    plt.colorbar(sc, ax=ax, shrink=0.85)
    ax.axhline(0, color='grey', lw=0.4, ls='--')
    ax.axvline(0, color='grey', lw=0.4, ls='--')
    ax.set_xlabel(f'PC{i+1} ({ev[i]:.1f}%)', fontsize=10)
    ax.set_ylabel(f'PC{j+1} ({ev[j]:.1f}%)', fontsize=10)
    ax.set_title(f'PC{i+1} vs PC{j+1}', fontsize=10)

fig.suptitle(
    f'PCA Dashboard — Normalised Totalmenge (no OdourType) | n={len(recipes_
    fontsize=12, fontweight='bold'
)
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_dashboard_pc123.png', dpi=150, bbox_inche
plt.show()
print('Saved: pca_v2_noOT_dashboard_pc123.png')
```



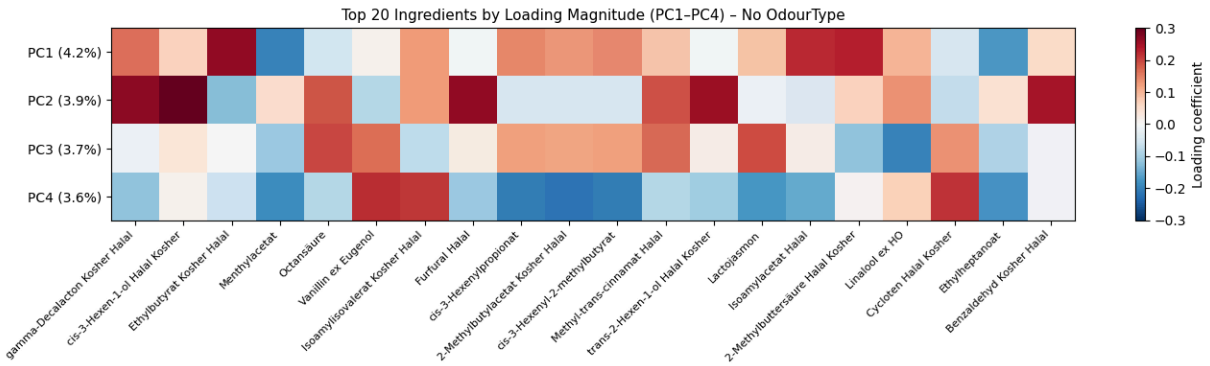

```
Saved: pca_v2_noOT_dashboard_pc123.png
```

6 · Loading Heatmap – Which Ingredients Drive Which Components

In [14]:

```
N_TOP = 20
N_PC  = min(4, pca_oav.n_components_)

importance_4 = np.sqrt((loadings[:N_PC]**2).sum(axis=0))
top20_idx    = np.argsort(importance_4)[::-1][:N_TOP]

top20_cas   = [cas_labels[i] for i in top20_idx]
top20_names = [cas_name.get(c, c).split(',')[0][:35] for c in top20_cas]
loading_mat = loadings[:N_PC][:, top20_idx]

fig, ax = plt.subplots(figsize=(14, 4))
im = ax.imshow(loading_mat, aspect='auto', cmap='RdBu_r', vmin=-0.3, vmax=0.
ax.set_xticks(range(N_TOP))
ax.set_xticklabels(top20_names, rotation=45, ha='right', fontsize=8)
ax.set_yticks(range(N_PC))
ax.set_yticklabels([f'PC{k+1} ({ev[k]:.1f}%)' for k in range(N_PC)], fontsiz
plt.colorbar(im, ax=ax, label='Loading coefficient')
ax.set_title(f'Top {N_TOP} Ingredients by Loading Magnitude (PC1–PC{N_PC}) -
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_loading_heatmap.png', dpi=150, bbox_inche
```

```
plt.show()
print('Saved: pca_v2_noOT_loading_heatmap.png')
```




```
Saved: pca_v2_noOT_loading_heatmap.png
```

7 · Recipe Scores Table

In [15]:
```
scores_df = pd.DataFrame(
    scores_oav[:, :5],
    index=pd.Index(recipes_oav, name='Rez.-Nr.'),
    columns=[f'PC{i+1}' for i in range(5)]
)
# Add Mahalanobis distance to centroid (PC1–PC2)
scores_df['Mahal_dist'] = dist_f.round(3)

print('PCA Scores (first 15 recipes):')
print(scores_df.head(15).round(3).to_string())

scores_df.to_csv(OUT_DIR / 'pca_v2_noOT_scores.csv')
print('\nFull table saved: pca_v2_noOT_scores.csv')
```

```
PCA Scores (first 15 recipes):
            PC1    PC2    PC3    PC4    PC5  Mahal_dist
Rez.-Nr.
185.046  -0.924  1.956 -0.628  0.183 -1.798       0.938
185.086  -0.931 -0.803  0.148  0.534 -0.467       0.526
185.090P -1.604 -1.051 -0.380 -0.027 -0.421       0.816
185.091  -6.495  0.522 -2.656 -5.055  4.134       2.740
185.133  -3.054 -1.687 -2.058 -2.023  1.264       1.480
185.237H -0.534 -0.477  0.353  0.641 -0.070       0.306
185.239  -0.935  1.663  3.200 -1.066 -1.211       0.826
185.267   0.045 -0.617 -1.141 -0.287 -0.684       0.270
185.291  -1.343 -0.522 -0.111  0.099 -0.391       0.609
185.294  -0.513 -0.564 -0.101  1.170  0.299       0.327
185.314  -0.717 -0.193 -0.786  0.748 -0.170       0.313
185.321   1.239  1.589  1.638 -1.027 -0.680       0.868
185.471   5.660 -3.639 -1.087 -2.170 -1.169       2.861
185.514  -7.555 -1.976 -5.427 -6.271  6.225       3.291
185.578P  2.705 -2.305  0.123  2.222  3.407       1.519

Full table saved: pca_v2_noOT_scores.csv
```

8 · Outlier Removal Summary

Complete log of which recipes were removed in which iteration and why.

In [16]:
```
print('=== Outlier Removal Summary ===')
print(f'Total iterations run  : {len(iteration_log)}')
print(f'Total recipes removed : {len(all_removed)}')
print(f'Starting recipes      : {len(pivot_oav)}')
print(f'Final clean recipes   : {len(recipes_oav)}')
print()

if all_removed:
    print(f'  {"Iteration":>9}  {"Recipe":>12}')
    print(f'  {"—"*9}  {"—"*12}')
    for it_n, rez in all_removed:
        print(f'  {it_n:>9}  {rez:>12}')
else:
    print('  No recipes were removed (data was already harmonious).')

print()
print('=== Per-Iteration Detail ===')
for row in iteration_log:
    outliers_str = ', '.join(row['outliers']) if row['outliers'] else 'none'
    print(f"  Iter {row['iteration']:>2}: {row['n_recipes']:>3} recipes  "
          f"PC1={row['ev_PC1']:>4.1f}%  PC2={row['ev_PC2']:>4.1f}%  "
          f"Outliers: {outliers_str}")
```

```
=== Outlier Removal Summary ===
Total iterations run  : 20
Total recipes removed : 43
Starting recipes      : 129
Final clean recipes   : 87
```

Iteration	Recipe
1	187.004P
1	185.309P
2	187.015P
2	187.482P
2	185.382P
3	188.462KHP
3	185.028
4	186.985P
4	185.748P
5	186.950P
5	186.490P
6	186.179P
6	186.395P
7	188.628P
7	187.507P
7	186.396P
8	187.894P
8	187.893P
8	187.871P
9	185.727P
9	187.796P
10	185.339P
10	185.507P
10	185.796P
11	187.061P
12	187.772
12	187.746P
12	185.043P
13	188.819
13	185.044
13	189.051
14	185.513P
14	186.569P
15	188.977
15	187.694P
16	187.662P
16	185.922P
17	188.992
17	187.695P
18	185.178HP
18	186.293P
19	185.902P
20	186.190P

```
=== Per-Iteration Detail ===
  Iter  1: 129 recipes  PC1= 5.2%  PC2= 4.8%  Outliers: 187.004P, 185.309P
  Iter  2: 127 recipes  PC1= 4.3%  PC2= 4.1%  Outliers: 187.015P, 187.482P,
185.382P
```

```
  Iter  3: 124 recipes  PC1= 4.0%  PC2= 3.7%  Outliers: 188.462KHP, 185.028
  Iter  4: 122 recipes  PC1= 4.1%  PC2= 3.9%  Outliers: 186.985P, 185.748P
  Iter  5: 120 recipes  PC1= 3.9%  PC2= 3.8%  Outliers: 186.950P, 186.490P
  Iter  6: 118 recipes  PC1= 4.1%  PC2= 3.7%  Outliers: 186.179P, 186.395P
  Iter  7: 116 recipes  PC1= 3.8%  PC2= 3.7%  Outliers: 188.628P, 187.507P,
186.396P
  Iter  8: 113 recipes  PC1= 3.9%  PC2= 3.5%  Outliers: 187.894P, 187.893P,
187.871P
  Iter  9: 110 recipes  PC1= 3.8%  PC2= 3.6%  Outliers: 185.727P, 187.796P
  Iter 10: 108 recipes  PC1= 3.9%  PC2= 3.4%  Outliers: 185.339P, 185.507P,
185.796P
  Iter 11: 105 recipes  PC1= 3.8%  PC2= 3.5%  Outliers: 187.061P
  Iter 12: 104 recipes  PC1= 3.8%  PC2= 3.5%  Outliers: 187.772, 187.746P, 1
85.043P
  Iter 13: 101 recipes  PC1= 4.2%  PC2= 3.6%  Outliers: 188.819, 185.044, 18
9.051
  Iter 14:  98 recipes  PC1= 3.8%  PC2= 3.7%  Outliers: 185.513P, 186.569P
  Iter 15:  96 recipes  PC1= 3.8%  PC2= 3.6%  Outliers: 188.977, 187.694P
  Iter 16:  94 recipes  PC1= 3.8%  PC2= 3.6%  Outliers: 187.662P, 185.922P
  Iter 17:  92 recipes  PC1= 4.0%  PC2= 3.7%  Outliers: 188.992, 187.695P
  Iter 18:  90 recipes  PC1= 4.1%  PC2= 3.9%  Outliers: 185.178HP, 186.293P
  Iter 19:  88 recipes  PC1= 4.1%  PC2= 4.0%  Outliers: 185.902P
  Iter 20:  87 recipes  PC1= 4.2%  PC2= 3.9%  Outliers: 186.190P
```

Summary

	Value
Version	v2 - without threshold, without OdourType
Pipeline	Remove ignore list → Normalise per recipe → Iterative PCA + outlier removal
Features	CAS-level normalised TotalIntensity
OdourType	Not used
Log transform	None
Outlier threshold	3σ Mahalanobis (2D PC1-PC2)
Max iterations	20
Outputs	`outputs/pca_v2_noOT_*`

In [17]:
```python
# ── Pre-compute full contribution tensor: (n_recipes, n_pc, n_cas) ──────
N_PC_ATTR = 4
contrib = X_oav_scaled[:, np.newaxis, :] * loadings[np.newaxis, :N_PC_ATTR,

recipe_idx_map = {r: idx for idx, r in enumerate(recipes_oav)}

print(f'Contribution tensor shape: {contrib.shape}  (recipes × PCs × ingredi
print('Ready for recipe lookup.')
```

```
Contribution tensor shape: (87, 4, 225)  (recipes × PCs × ingredients)
Ready for recipe lookup.
```

In [18]:

```python
def explain_recipe(rez_nr, n_top=8, plot=True):
    if rez_nr not in recipe_idx_map:
        print(f'Recipe {rez_nr!r} not found in clean dataset.')
        print('Available (first 10):', sorted(recipe_idx_map.keys())[:10])
        return

    r_idx    = recipe_idx_map[rez_nr]
    r_scores = scores_oav[r_idx, :N_PC_ATTR]
    r_contrib = contrib[r_idx]

    print('=' * 50)
    print(f'  Recipe: {rez_nr}')
    print('=' * 50)
    for k in range(N_PC_ATTR):
        print(f'  PC{k+1} score: {r_scores[k]:+.3f}  (explains {ev[k]:.1f}%)
    print()

    for k in range(N_PC_ATTR):
        c_vec     = r_contrib[k]
        top_idx_k = np.argsort(np.abs(c_vec))[::-1][:n_top]
        print(f'  -- PC{k+1}: top {n_top} ingredient drivers --')
        for i in top_idx_k:
            cas  = cas_labels[i]
            name = cas_name.get(cas, cas).split(',')[0][:40]
            val  = final_pivot.iloc[r_idx, i]
            sign = '+' if c_vec[i] >= 0 else '-'
            print(f'    {sign}  contrib={c_vec[i]:+.4f}  norm={val:.6f}  '
                  f'{cas:15s}  {name}')
        print()

    dists  = np.linalg.norm(scores_oav - r_scores, axis=1)
    nn_idx = np.argsort(dists)[1:9]
    print('  -- 8 nearest recipes --')
    for ri in nn_idx:
        print(f'    {recipes_oav[ri]:12s}  dist={dists[ri]:.3f}'
              f'  PC1={scores_oav[ri,0]:+.2f}  PC2={scores_oav[ri,1]:+.2f}')

    if plot:
        fig, ax = plt.subplots(figsize=(10, 7))
        ax.scatter(scores_oav[:, 0], scores_oav[:, 1],
                   c='#4A90D9', s=40, alpha=0.4, edgecolors='white', linewic
        ax.scatter(scores_oav[r_idx, 0], scores_oav[r_idx, 1],
                   c='#E05A2B', s=220, zorder=5, edgecolors='#8B0000', linew
                   label=str(rez_nr))
        for ri in nn_idx:
            ax.scatter(scores_oav[ri, 0], scores_oav[ri, 1],
                       c='#27AE60', s=90, zorder=4, edgecolors='white', line
        ax.axhline(0, color='grey', lw=0.5, ls='--')
        ax.axvline(0, color='grey', lw=0.5, ls='--')
        ax.set_xlabel(f'PC1 ({ev[0]:.1f}%)')
        ax.set_ylabel(f'PC2 ({ev[1]:.1f}%)')
        ax.set_title(f'Recipe {rez_nr} in PCA Space (red) + 8 nearest (greer
        ax.legend()
```

```
        plt.tight_layout()
        plt.show()
```

---

9 · Ingredient → Recipe Lookup

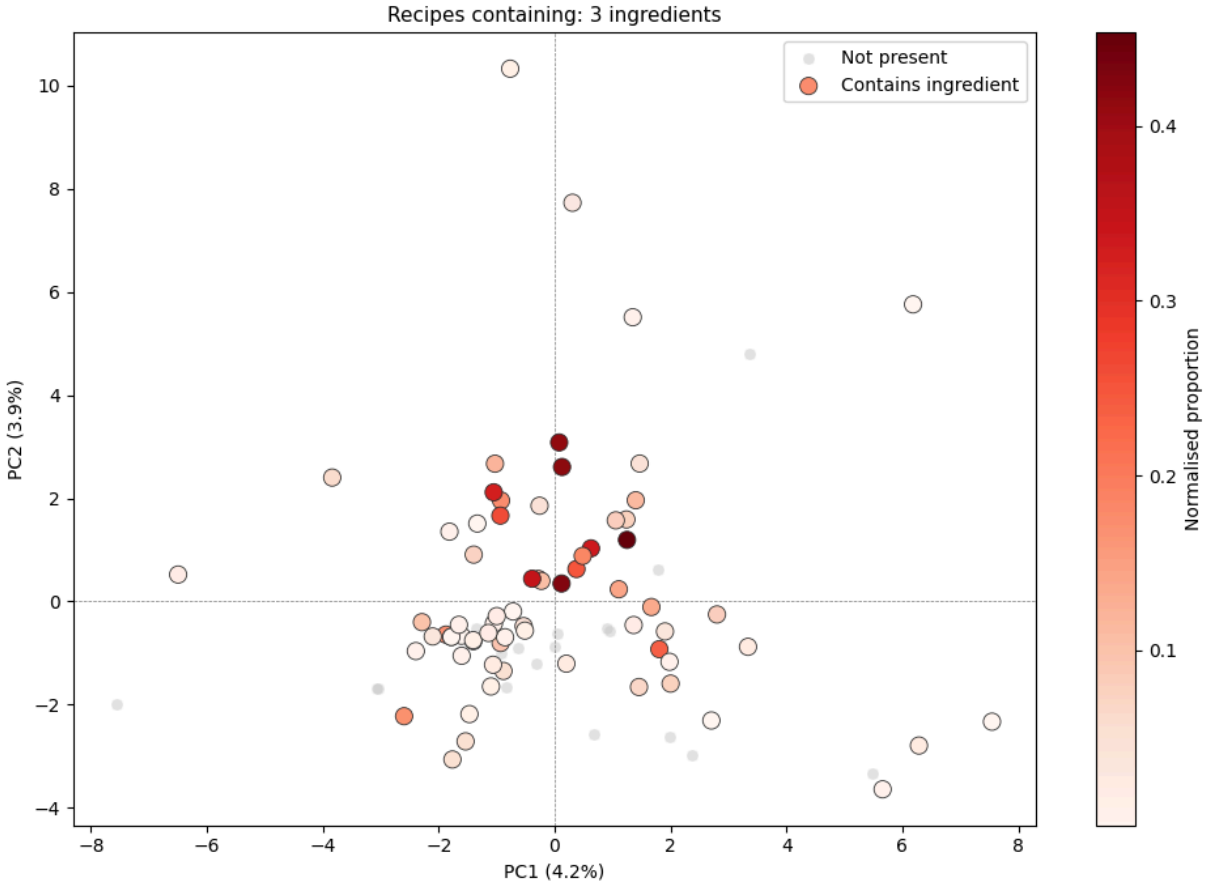
Search by CAS number or name fragment (case-insensitive). Recipes are ranked by normalised proportion for that ingredient.

In [19]:
```
def find_by_ingredient(query, n_top=15, plot=True):
    q = str(query).strip()
    if q in cas_labels:
        matched_cas = [q]
    else:
        matched_cas = [c for c in cas_labels if q.lower() in cas_name.get(c,

    if not matched_cas:
        print(f'No ingredient found matching {q!r}.')
        return None

    print(f'Matched {len(matched_cas)} CAS number(s):')
    for cas in matched_cas:
        print(f'  {cas}  -->  {cas_name.get(cas, cas)}')
    print()

    norm_per_recipe = final_pivot[matched_cas].sum(axis=1)
    nonzero         = norm_per_recipe[norm_per_recipe > 0].sort_values(ascer
    print(f'Present in {len(nonzero)} / {len(recipes_oav)} recipes (clean da
    print()

    table_rows = []
    for rez, norm_val in nonzero.head(n_top).items():
        ri = recipe_idx_map[rez]
        table_rows.append({
            'Rez.-Nr.':   rez,
            'Norm_Total': round(norm_val, 6),
            'PC1':        round(scores_oav[ri, 0], 3),
            'PC2':        round(scores_oav[ri, 1], 3),
            'PC3':        round(scores_oav[ri, 2], 3),
        })

    result = pd.DataFrame(table_rows).set_index('Rez.-Nr.')
    print(result.to_string())
    print()

    if plot:
        fig, ax = plt.subplots(figsize=(10, 7))
        match_mask = norm_per_recipe.reindex(recipes_oav, fill_value=0).valu
        ax.scatter(scores_oav[~match_mask, 0], scores_oav[~match_mask, 1],
                   c='#BBBBBB', s=40, alpha=0.4, edgecolors='white', linewic
                   label='Not present')
        match_vals = norm_per_recipe.reindex(recipes_oav, fill_value=0).valu
        sc = ax.scatter(scores_oav[match_mask, 0], scores_oav[match_mask, 1]
```

```
                        c=match_vals, cmap='Reds',
                        s=90, zorder=4, edgecolors='#333', linewidths=0.6,
                        label='Contains ingredient')
        plt.colorbar(sc, ax=ax, label='Normalised proportion')
        ax.axhline(0, color='grey', lw=0.5, ls='--')
        ax.axvline(0, color='grey', lw=0.5, ls='--')
        ax.set_xlabel(f'PC1 ({ev[0]:.1f}%)')
        ax.set_ylabel(f'PC2 ({ev[1]:.1f}%)')
        name_str = cas_name.get(matched_cas[0], matched_cas[0]).split(',')[0
            if len(matched_cas) == 1 else f'{len(matched_cas)} ingredients'
        ax.set_title(f'Recipes containing: {name_str}', fontsize=11)
        ax.legend()
        plt.tight_layout()
        plt.show()

    return result
```

In [20]:
```
# ── Demo: Furaneol — the key strawberry character odorant ──────────────
_ = find_by_ingredient('furaneol', n_top=15, plot=True)
```

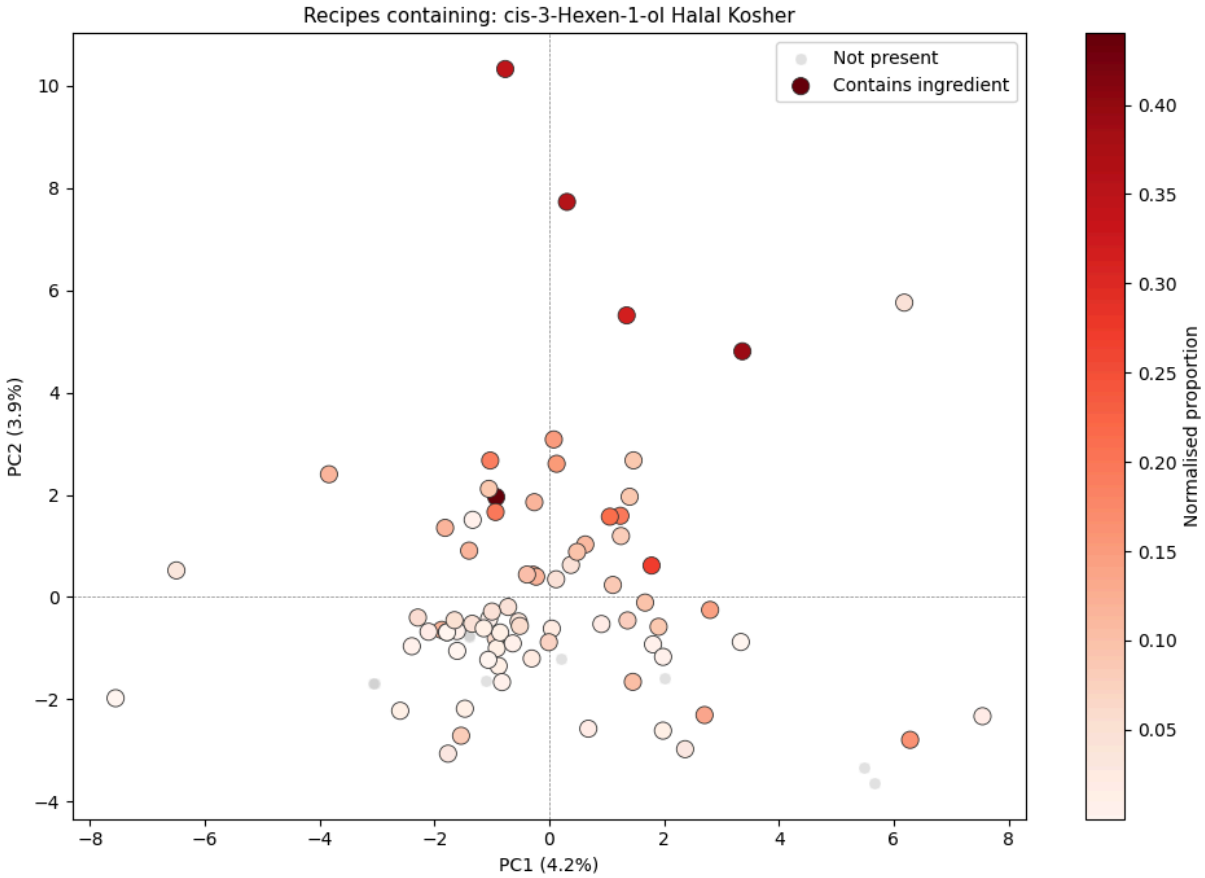
```
Matched 3 CAS number(s):
  27538-09-6   -->  Ethylfuraneol Kosher Halal
  3658-77-3  -->  Furaneol Halal
  4166-20-5  -->  Furaneolacetat Halal Kosher

Present in 69 / 87 recipes (clean dataset).

          Norm_Total    PC1    PC2    PC3
Rez.-Nr.
187.799P    0.453288  1.249  1.194 -1.704
188.491P    0.428472  0.119  0.348 -0.206
187.753P    0.415469  0.128  2.606  0.525
187.752P    0.411091  0.079  3.081  0.594
188.822     0.347201 -0.388  0.441  0.264
187.567P    0.336199  0.628  1.029  0.562
185.664     0.327588 -1.051  2.116  3.476
185.239     0.261752 -0.935  1.663  3.200
187.365     0.251546  0.377  0.629 -0.092
187.064P    0.234428  1.808 -0.924 -0.584
187.608P    0.185825  0.485  0.883  0.668
185.046     0.176817 -0.924  1.956 -0.628
188.639P    0.171056 -2.597 -2.222  5.122
187.776     0.166385 -1.874 -0.641 -0.472
189.075     0.145064  1.108  0.237  0.040
```



In [21]:
```
# ── Demo: top PC2 driver ingredient ──────────────────────────────────
top_pc2_cas = cas_labels[np.argmax(np.abs(loadings[1]))]
print(f'Top PC2 driver: {top_pc2_cas} — {cas_name.get(top_pc2_cas, "?")}')
print()
_ = find_by_ingredient(top_pc2_cas, n_top=15, plot=True)
```

```
Top PC2 driver: 928-96-1 — cis-3-Hexen-1-ol Halal Kosher

Matched 1 CAS number(s):
  928-96-1  -->  cis-3-Hexen-1-ol Halal Kosher

Present in 77 / 87 recipes (clean dataset).
```

Rez.-Nr.	Norm_Total	PC1	PC2	PC3
185.046	0.440079	-0.924	1.956	-0.628
188.412P	0.392468	3.365	4.804	-4.751
186.192	0.357089	0.308	7.725	0.457
186.190P	0.344461	-0.766	10.322	0.972
187.519P	0.313607	1.348	5.507	2.759
187.657P	0.272330	1.782	0.617	-1.664
188.820	0.214821	1.056	1.572	-0.007
185.239	0.197980	-0.935	1.663	3.200
185.321	0.195215	1.239	1.589	1.638
188.894	0.190807	-1.028	2.670	4.974
187.483P	0.161988	6.287	-2.790	1.216
187.753P	0.154424	0.128	2.606	0.525
187.752P	0.152797	0.079	3.081	0.594
187.897P	0.141941	2.802	-0.251	0.971
185.578P	0.135751	2.705	-2.305	0.123



---

10 · Virtual Recipe → Nearest Real Recipes

Specify any combination of ingredients and amounts → projected into PCA space → returns the closest real recipes from the clean dataset.

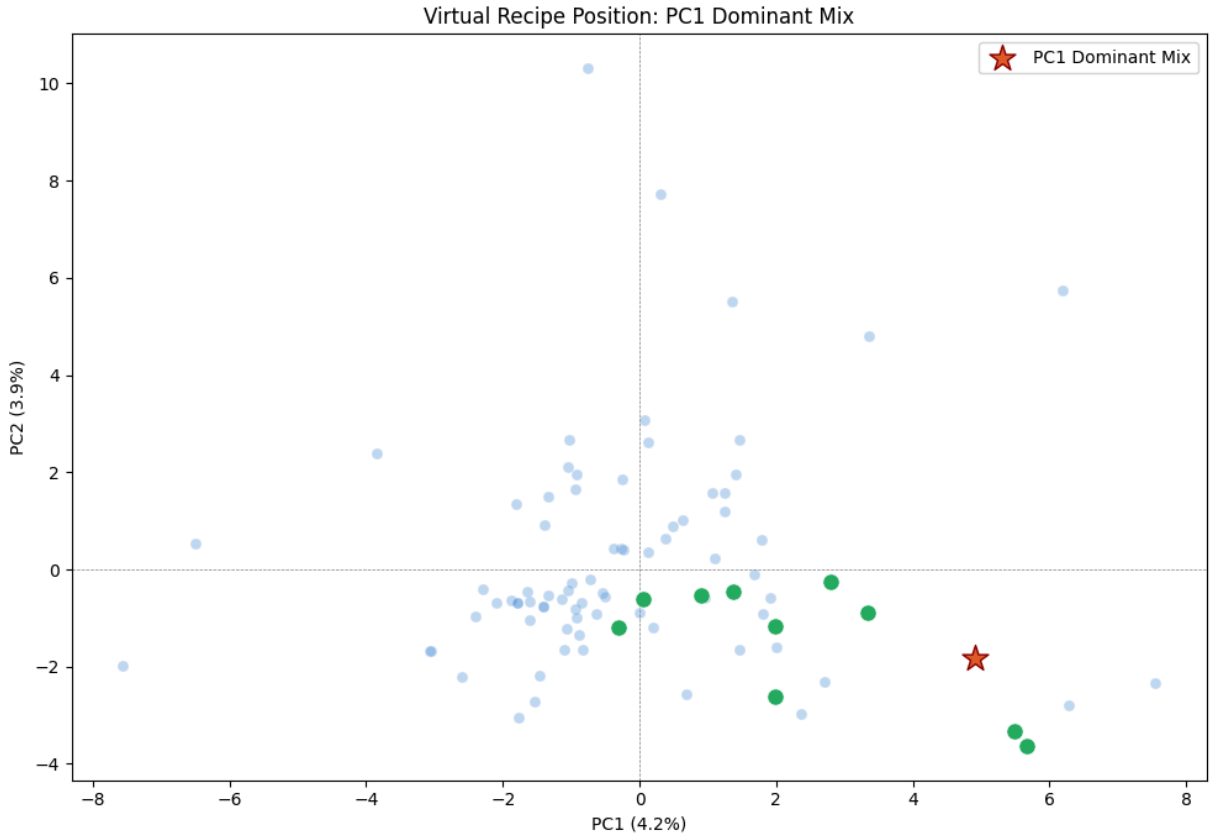
In [22]:
```
def nearest_recipes(ingredient_dict, n=10, plot=True, label='Virtual Recipe'
    resolved = {}
    for key, amt in ingredient_dict.items():
        if key in cas_labels:
            resolved[key] = amt
        else:
            hit = next(
                (c for c in cas_labels if key.lower() in cas_name.get(c, '')
                None
            )
            if hit:
                resolved[hit] = amt
            else:
                print(f'  WARNING: {key!r} not found in model - skipped.')

    if not resolved:
        print('No valid ingredients after resolution. Aborting.')
        return None

    print(f'  {label}  -  resolved {len(resolved)} ingredient(s):')
    for cas, amt in resolved.items():
        print(f'    {cas:15s}  {cas_name.get(cas, cas).split(",")[0][:40]:40
    print()

    virt_vec = np.zeros(len(cas_labels))
    for cas, amt in resolved.items():
        virt_vec[cas_labels.index(cas)] = max(0.0, float(amt))
    if virt_vec.sum() > 0:
        virt_vec /= virt_vec.sum()

    virt_scaled = scaler_oav.transform(virt_vec.reshape(1, -1))[0]
    virt_score  = pca_oav.transform(virt_scaled.reshape(1, -1))[0]

    print(f'  Virtual recipe in PCA space: '
          f'PC1={virt_score[0]:+.3f}  PC2={virt_score[1]:+.3f}  PC3={virt_sc
    print()

    dists  = np.linalg.norm(scores_oav - virt_score, axis=1)
    nn_idx = np.argsort(dists)[:n]

    result_rows = []
    print(f'  {n} nearest real recipes:')
    for rank, ri in enumerate(nn_idx, 1):
        result_rows.append({
            'Rank':     rank,
            'Rez.-Nr.': recipes_oav[ri],
            'Distance': round(dists[ri], 3),
            'PC1':      round(scores_oav[ri, 0], 3),
            'PC2':      round(scores_oav[ri, 1], 3),
            'PC3':      round(scores_oav[ri, 2], 3),
        })
```

```
        print(f'    #{rank:>2}  {recipes_oav[ri]:12s}  dist={dists[ri]:.3f}'
              f'  PC1={scores_oav[ri,0]:+.2f}  PC2={scores_oav[ri,1]:+.2f}')

    if plot:
        fig, ax = plt.subplots(figsize=(10, 7))
        ax.scatter(scores_oav[:, 0], scores_oav[:, 1],
                   c='#4A90D9', s=40, alpha=0.35, edgecolors='white', linewi
        for ri in nn_idx:
            ax.scatter(scores_oav[ri, 0], scores_oav[ri, 1],
                       c='#27AE60', s=90, zorder=4, edgecolors='white', line
        ax.scatter(virt_score[0], virt_score[1],
                   c='#E05A2B', s=250, marker='*', zorder=5,
                   edgecolors='#8B0000', linewidths=1.0, label=label)
        ax.axhline(0, color='grey', lw=0.5, ls='--')
        ax.axvline(0, color='grey', lw=0.5, ls='--')
        ax.set_xlabel(f'PC1 ({ev[0]:.1f}%)')
        ax.set_ylabel(f'PC2 ({ev[1]:.1f}%)')
        ax.set_title(f'Virtual Recipe Position: {label}')
        ax.legend()
        plt.tight_layout()
        plt.show()

    return pd.DataFrame(result_rows).set_index('Rank')
```

In [23]:

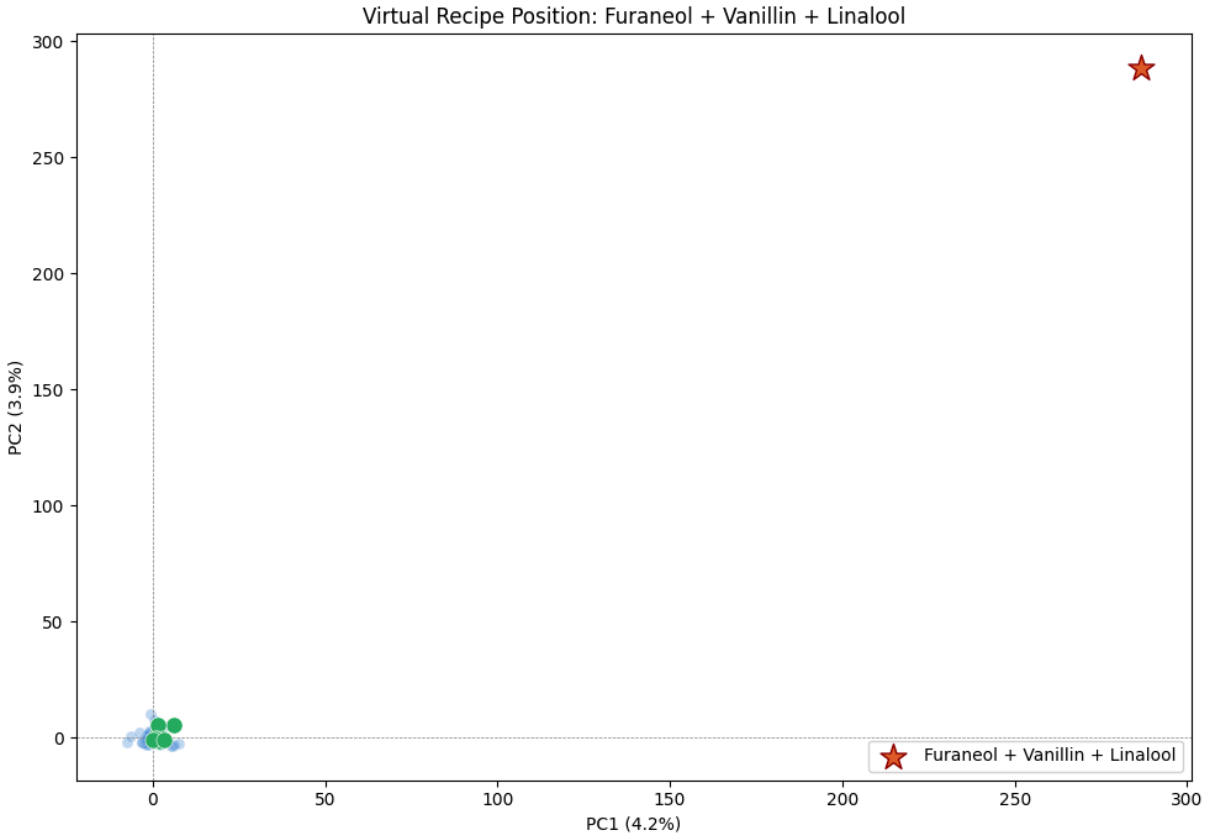
```
# -- Demo 1: auto-build virtual recipe from top PC1 ingredients ------------
top3_pc1_idx = np.argsort(np.abs(loadings[0]))[::-1][:3]
top3_pc1_cas = [cas_labels[i] for i in top3_pc1_idx]

pc1_mix = {}
for cas in top3_pc1_cas:
    if cas not in final_pivot.columns:
        continue
    vals = final_pivot.loc[final_pivot[cas] > 0, cas]
    if len(vals) == 0:
        continue
    pc1_mix[cas] = float(vals.median())

print('Demo virtual recipe (top-3 PC1 ingredients at median normalised propo
for cas, amt in pc1_mix.items():
    print(f'  {cas}  {cas_name.get(cas, "?").split(",")[0][:40]}  norm={amt:
print()
_ = nearest_recipes(pc1_mix, n=10, label='PC1 Dominant Mix')
```

```
Demo virtual recipe (top-3 PC1 ingredients at median normalised proportion):
  105-54-4  Ethylbutyrat Kosher Halal  norm=0.077262
  116-53-0  2-Methylbuttersäure Halal Kosher  norm=0.051147
  123-92-2  Isoamylacetat Halal  norm=0.008833

  PC1 Dominant Mix  -  resolved 3 ingredient(s):
    105-54-4         Ethylbutyrat Kosher Halal                 norm=0.077262
    116-53-0         2-Methylbuttersäure Halal Kosher          norm=0.051147
    123-92-2         Isoamylacetat Halal                       norm=0.008833

  Virtual recipe in PCA space: PC1=+4.913  PC2=-1.827  PC3=-0.656

  10 nearest real recipes:
    # 1  187.897P      dist=3.901  PC1=+2.80  PC2=-0.25
    # 2  187.246P      dist=4.291  PC1=+1.98  PC2=-1.17
    # 3  187.791P      dist=5.713  PC1=+1.98  PC2=-2.61
    # 4  187.167P      dist=5.812  PC1=+5.48  PC2=-3.33
    # 5  188.560P      dist=5.872  PC1=+0.91  PC2=-0.53
    # 6  185.267       dist=6.010  PC1=+0.04  PC2=-0.62
    # 7  185.471       dist=6.167  PC1=+5.66  PC2=-3.64
    # 8  188.118P      dist=6.274  PC1=+1.36  PC2=-0.46
    # 9  187.414P      dist=6.292  PC1=-0.31  PC2=-1.20
    #10  185.984P      dist=6.318  PC1=+3.34  PC2=-0.88
```



In [24]:
```
# -- Demo 2: freely-defined mix (edit to explore) ---------------------------
my_mix = {
    'furaneol':  0.30,
    'vanillin':  0.10,
    'linalool':  0.05,
}
_ = nearest_recipes(my_mix, n=10, label='Furaneol + Vanillin + Linalool')
```

```
Furaneol + Vanillin + Linalool  -  resolved 3 ingredient(s):
  27538-09-6       Ethylfuraneol Kosher Halal                norm=0.300000
  121-33-5         Vanillin ex Eugenol                       norm=0.100000
  1365-19-1        Linalooloxid                              norm=0.050000

Virtual recipe in PCA space: PC1=+286.540  PC2=+288.598  PC3=-377.886

10 nearest real recipes:
  # 1  186.981P      dist=1055.422  PC1=+6.18  PC2=+5.76
  # 2  189.012       dist=1071.130  PC1=+1.40  PC2=+1.96
  # 3  187.799P      dist=1072.345  PC1=+1.25  PC2=+1.19
  # 4  187.519P      dist=1072.393  PC1=+1.35  PC2=+5.51
  # 5  187.786P      dist=1072.685  PC1=+0.96  PC2=-0.58
  # 6  188.740P      dist=1073.046  PC1=+2.00  PC2=-1.59
  # 7  187.897P      dist=1073.352  PC1=+2.80  PC2=-0.25
  # 8  188.560P      dist=1073.523  PC1=+0.91  PC2=-0.53
  # 9  185.267       dist=1073.545  PC1=+0.04  PC2=-0.62
  #10  185.984P      dist=1073.626  PC1=+3.34  PC2=-0.88
```

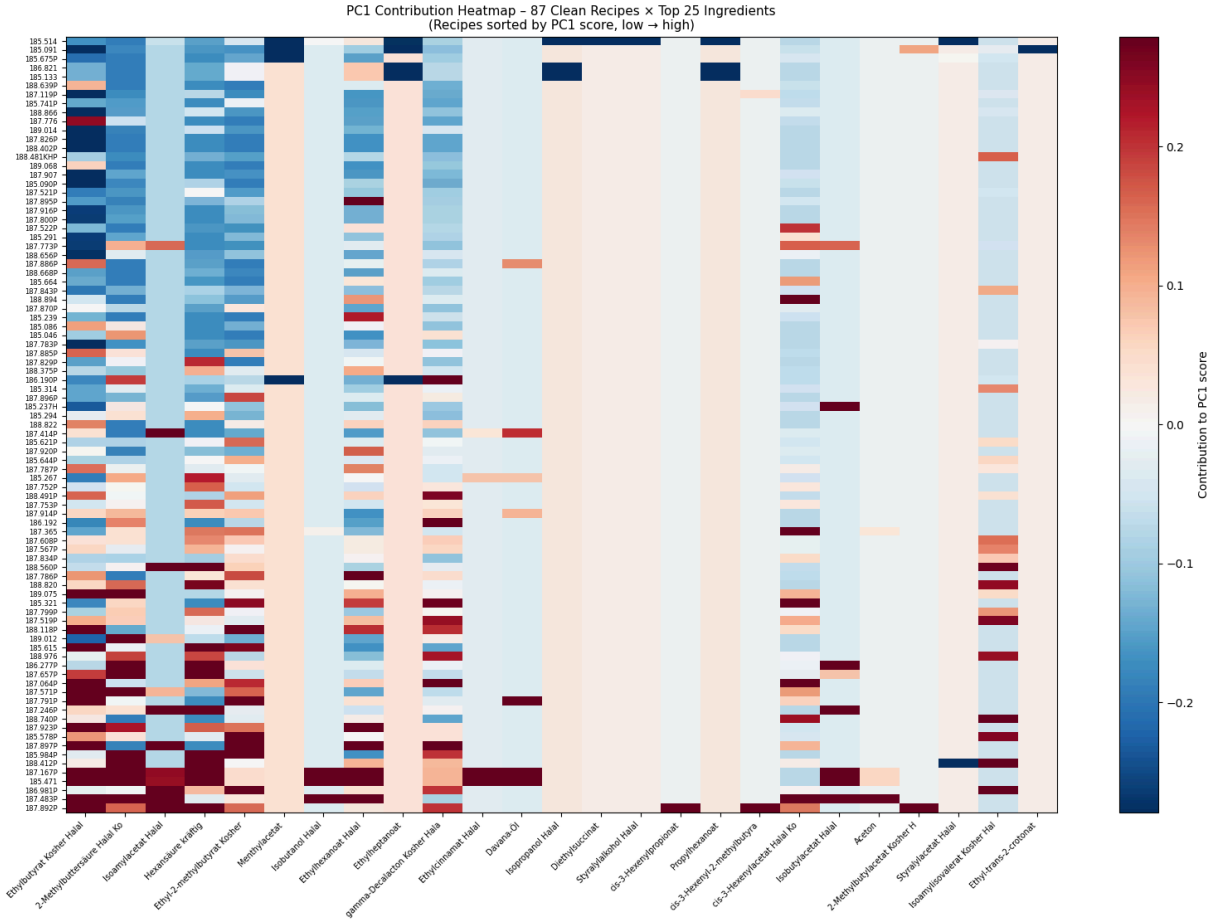


---

11 · Full Contribution Heatmap + Attribution Export

In [25]:

```
# ── PC1 contribution heatmap — all clean recipes × top-25 ingredients ─────
N_ING_HEAT = 25

top25_idx = np.argsort(np.abs(loadings[0]))[::-1][:N_ING_HEAT]
top25_cas = [cas_labels[i] for i in top25_idx]
top25_lbl = [cas_name.get(c, c).split(',')[0][:28] for c in top25_cas]

contrib_pc1     = contrib[:, 0, :][:, top25_idx]
order           = np.argsort(scores_oav[:, 0])
contrib_sorted  = contrib_pc1[order]
recipe_lbl_sort = [recipes_oav[i] for i in order]

fig, ax = plt.subplots(figsize=(16, max(8, len(recipes_oav) * 0.13)))
vmax = np.percentile(np.abs(contrib_sorted), 95)
im = ax.imshow(contrib_sorted, aspect='auto', cmap='RdBu_r',
               vmin=-vmax, vmax=vmax, interpolation='nearest')
ax.set_xticks(range(N_ING_HEAT))
ax.set_xticklabels(top25_lbl, rotation=45, ha='right', fontsize=7)
ax.set_yticks(range(len(recipes_oav)))
ax.set_yticklabels(recipe_lbl_sort, fontsize=6)
plt.colorbar(im, ax=ax, label='Contribution to PC1 score')
ax.set_title(
    f'PC1 Contribution Heatmap — {len(recipes_oav)} Clean Recipes × Top {N_I
    '(Recipes sorted by PC1 score, low → high)',
    fontsize=11
)
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_contrib_heatmap_pc1.png', dpi=150, bbox_i
plt.show()
print('Saved: pca_v2_noOT_contrib_heatmap_pc1.png')
```



Saved: pca_v2_noOT_contrib_heatmap_pc1.png

In [26]:
```
# ── Export: per-recipe top-5 ingredient driver per PC → CSV ──────────
export_rows = []
for r_idx, rez in enumerate(recipes_oav):
    row = {'Rez.-Nr.': rez}
    for k in range(N_PC_ATTR):
        c_vec = contrib[r_idx, k]
        top5  = np.argsort(np.abs(c_vec))[::-1][:5]
        row[f'PC{k+1}_score'] = round(scores_oav[r_idx, k], 4)
        for rank, i in enumerate(top5, 1):
            cas  = cas_labels[i]
            name = cas_name.get(cas, cas).split(',')[0][:35]
            row[f'PC{k+1}_top{rank}_cas']    = cas
            row[f'PC{k+1}_top{rank}_name']   = name
            row[f'PC{k+1}_top{rank}_contrib'] = round(float(c_vec[i]), 5)
    export_rows.append(row)

attr_df = pd.DataFrame(export_rows).set_index('Rez.-Nr.')
attr_df.to_csv(OUT_DIR / 'pca_v2_noOT_recipe_attribution.csv')
print('Exported: pca_v2_noOT_recipe_attribution.csv')
print(f'Shape: {attr_df.shape}')
```

```
Exported: pca_v2_noOT_recipe_attribution.csv
Shape: (87, 64)
```

12 · Top 20 Ingredients per Recipe (PCA Contribution Export)

For every recipe the 20 ingredients that contribute most to its PCA fingerprint. Only ingredients actually present in the recipe (Norm_Totalmenge > 0) are included.

In [27]:
```
total_abs_contrib = np.abs(contrib).sum(axis=1)  # (n_recipes, n_cas)
N_TOP_PER_RECIPE  = 20
rows = []

for r_idx, rez in enumerate(recipes_oav):
    present_mask   = final_pivot.iloc[r_idx].values > 0
    present_idx    = np.where(present_mask)[0]
    if len(present_idx) == 0:
        continue
    contribs_pres  = total_abs_contrib[r_idx][present_idx]
    top_k          = min(N_TOP_PER_RECIPE, len(present_idx))
    top_local      = np.argsort(contribs_pres)[::-1][:top_k]
    top_idx_abs    = present_idx[top_local]

    for rank, i in enumerate(top_idx_abs, 1):
        cas = cas_labels[i]
        rows.append({
            'Rez.-Nr.':        rez,
            'Rank':            rank,
            'CAS':             cas,
            'Ingredient':      cas_name.get(cas, cas),
            'Norm_Totalmenge': round(float(final_pivot.iloc[r_idx, i]), 6),
            'Total_Contrib':   round(float(total_abs_contrib[r_idx, i]), 5),
            'PC1_Contrib':     round(float(contrib[r_idx, 0, i]), 5),
            'PC2_Contrib':     round(float(contrib[r_idx, 1, i]), 5),
            'PC3_Contrib':     round(float(contrib[r_idx, 2, i]), 5),
            'PC4_Contrib':     round(float(contrib[r_idx, 3, i]), 5),
        })

top20_df = pd.DataFrame(rows)
print(f'Shape: {top20_df.shape}')
print(f'Recipes covered: {top20_df["Rez.-Nr."].nunique()}')
print(f'All Norm_Totalmenge > 0: {(top20_df["Norm_Totalmenge"] > 0).all()}')
print()
print(top20_df.head(20).to_string(index=False))

top20_df.to_csv(OUT_DIR / 'pca_v2_noOT_top20_per_recipe.csv', index=False)
top20_df.to_excel(OUT_DIR / 'pca_v2_noOT_top20_per_recipe.xlsx', index=False
print('\nExported: pca_v2_noOT_top20_per_recipe.csv / .xlsx')
```

```
Shape: (1533, 10)
Recipes covered: 87
All Norm_Totalmenge > 0: True

Rez.-Nr.  Rank        CAS                          Ingredient  Norm_Totalmeng
e  Total_Contrib  PC1_Contrib  PC2_Contrib  PC3_Contrib  PC4_Contrib
 185.046     1   928-96-1      cis-3-Hexen-1-ol Halal Kosher         0.44007
9        1.61221      0.25127      1.18959      0.12627      0.04509
 185.046     2   110-43-0                  2-Heptanon Halal         0.01964
6        0.70537     -0.16510      0.37258     -0.12752      0.04016
 185.046     3   109-21-7                Butylbutyrat Halal         0.14538
3        0.69960     -0.16330      0.37311     -0.12528      0.03790
 185.046     4  8022-96-6      Jasmin-Absolue (blumig) Halal         0.00196
5        0.38261      0.13791     -0.01154     -0.19713     -0.03603
 185.046     5  3658-77-3                   Furaneol Halal         0.17681
7        0.26217      0.00068      0.09548      0.07253      0.09348
 185.046     6   116-53-0   2-Methylbuttersäure Halal Kosher         0.08644
4        0.22500      0.12432      0.03433     -0.06181      0.00454
 185.046     7   105-54-4          Ethylbutyrat Kosher Halal         0.07072
7        0.16177     -0.09475      0.04518     -0.00037      0.02147
 185.046     8   706-14-9      gamma-Decalacton Kosher Halal         0.04322
2        0.13757      0.04016      0.06550     -0.00326     -0.02865
 185.046     9   118-71-8               Maltol Kosher Halal         0.01571
7        0.02221     -0.00377     -0.01094     -0.00130     -0.00619
 185.086     1   106-32-1         Ethyloctanoat Halal Kosher         0.00375
6        0.69604      0.11234      0.17926     -0.18633      0.21810
 185.086     2   600-14-6              Acetylpropionyl Halal         0.03262
4        0.56449     -0.21629     -0.23205      0.03846      0.07770
 185.086     3 68917-33-9       Zitronenterpene Kosher Halal         0.00300
5        0.37817      0.11273     -0.12731      0.00872      0.12941
 185.086     4   706-14-9      gamma-Decalacton Kosher Halal         0.00901
5        0.36708     -0.10716     -0.17477      0.00871      0.07645
 185.086     5 16491-36-4         cis-3-Hexenylbutyrat Halal         0.01223
4        0.33324     -0.03834     -0.01935      0.19148      0.08407
 185.086     6   513-86-0     Acetoin 50% in PG Kosher Halal         0.04399
9        0.27385     -0.09073      0.00096      0.17326     -0.00890
 185.086     7  7452-79-1 Ethyl-2-methylbutyrat Kosher Halal         0.01330
7        0.24667     -0.13272      0.02627      0.06886      0.01882
 185.086     8   105-54-4          Ethylbutyrat Kosher Halal         0.13414
4        0.19287      0.11296     -0.05386      0.00045     -0.02560
 185.086     9   141-78-6           Ethylacetat Kosher Halal         0.00171
7        0.17934     -0.05474      0.02365     -0.08614      0.01480
 185.086    10   928-96-1      cis-3-Hexen-1-ol Halal Kosher         0.05838
0        0.07856     -0.01224     -0.05797     -0.00615     -0.00220
 185.086    11    66-25-1               Hexanal Kosher Halal         0.00032
2        0.06096      0.02947      0.00951      0.01114      0.01084

Exported: pca_v2_noOT_top20_per_recipe.csv / .xlsx
```

---

13 · Top 20 Ingredients Globally (Across All Clean Recipes)

In [28]:

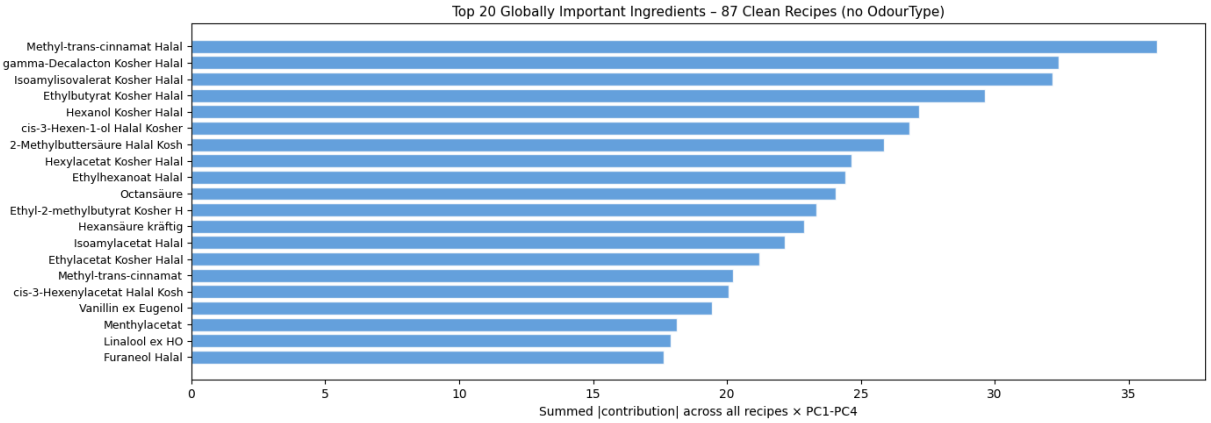
```
global_importance    = np.abs(contrib).sum(axis=(0, 1))  # (n_cas,)
top20_global_idx     = np.argsort(global_importance)[::-1][:20]

global_rows = []
for rank, i in enumerate(top20_global_idx, 1):
    cas = cas_labels[i]
    col = final_pivot.iloc[:, i]
    global_rows.append({
        'Rank':              rank,
        'CAS':               cas,
        'Ingredient':        cas_name.get(cas, cas),
        'Global_Importance': round(float(global_importance[i]), 4),
        'Frequency':         round(float((col > 0).mean()), 4),
        'Recipes_Count':     int((col > 0).sum()),
        'Avg_Norm_Present':  round(float(col[col > 0].mean() if (col > 0).ar
        'PC1_Loading':       round(float(loadings[0, i]), 4),
        'PC2_Loading':       round(float(loadings[1, i]), 4),
        'PC3_Loading':       round(float(loadings[2, i]), 4),
        'PC4_Loading':       round(float(loadings[3, i]), 4),
    })

top20_global_df = pd.DataFrame(global_rows)
print('=== Top 20 Ingredients Globally ===')
print(top20_global_df.to_string(index=False))

top20_global_df.to_csv(OUT_DIR / 'pca_v2_noOT_top20_ingredients_global.csv',
top20_global_df.to_excel(OUT_DIR / 'pca_v2_noOT_top20_ingredients_global.xls
print('\nExported: pca_v2_noOT_top20_ingredients_global.csv / .xlsx')

names_short = [cas_name.get(cas_labels[i], cas_labels[i]).split(',')[0][:30]
               for i in top20_global_idx]
fig, ax = plt.subplots(figsize=(14, 5))
ax.barh(range(20), global_importance[top20_global_idx[::-1]],
        color='#4A90D9', alpha=0.85, edgecolor='white')
ax.set_yticks(range(20))
ax.set_yticklabels(names_short[::-1], fontsize=9)
ax.set_xlabel('Summed |contribution| across all recipes × PC1–PC4', fontsize
ax.set_title(
    f'Top 20 Globally Important Ingredients — {len(recipes_oav)} Clean Recip
    fontsize=11
)
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_top20_global_bar.png', dpi=150, bbox_inch
plt.show()
print('Saved: pca_v2_noOT_top20_global_bar.png')
```

=== Top 20 Ingredients Globally ===

Rank	CAS	Ingredient	Global_Importance	Frequency	Recipes_Count	Avg_Norm_Present	PC1_Loading	PC2_Loading	PC3_Loading	PC4_Loading
1	1754-62-7	Methyl-trans-cinnamat Halal	36.0639	0.4598	40	0.058822	0.0799	0.1904	0.1680	-0.0843
2	706-14-9	gamma-Decalacton Kosher Halal	32.4010	0.9540	83	0.035530	0.1628	0.2655	-0.0132	-0.1161
3	659-70-1	Isoamylisovalerat Kosher Halal	32.1475	0.3218	28	0.007990	0.1244	0.1263	-0.0756	0.2111
4	105-54-4	Ethylbutyrat Kosher Halal	29.6257	1.0000	87	0.099655	0.2634	-0.1256	0.0010	-0.0597
5	111-27-3	Hexanol Kosher Halal	27.1897	0.3563	31	0.014861	0.0445	0.1389	-0.0878	0.1861
6	928-96-1	cis-3-Hexen-1-ol Halal Kosher	26.8177	0.8851	77	0.086000	0.0654	0.3096	0.0329	0.0117
7	116-53-0	2-Methylbuttersäure Halal Kosher	25.8703	0.8161	71	0.063775	0.2333	0.0644	-0.1160	0.0085
8	142-92-7	Hexylacetat Kosher Halal	24.6693	0.4023	35	0.008380	0.0525	0.0867	0.1140	0.1653
9	123-66-0	Ethylhexanoat Halal	24.4166	0.9425	82	0.037652	0.1853	-0.0951	0.0264	-0.0994
10	124-07-2	Octansäure	24.0573	0.1264	11	0.010036	-0.0524	0.1855	0.1986	-0.0835
11	7452-79-1	Ethyl-2-methylbutyrat Kosher Halal	23.3353	0.9080	79	0.049158	0.2161	-0.0428	-0.1121	-0.0307
12	142-62-1	Hexansäure kräftig	22.8921	0.7471	65	0.019203	0.2241	-0.0146	-0.1066	0.0039
13	123-92-2	Isoamylacetat Halal	22.1553	0.1494	13	0.009434	0.2245	-0.0406	0.0176	-0.1505
14	141-78-6	Ethylacetat Kosher Halal	21.2214	0.6552	57	0.035971	0.1214	-0.0524	0.1910	-0.0328
15	103-26-4	Methyl-trans-cinnamat, natürlich Kosher Halal	20.2135	0.4138	36	0.033006	0.0747	-0.1091	-0.1354	-0.0147
16	3681-71-8	cis-3-Hexenylacetat Halal Kosher	20.0592	0.7011	61	0.011042	0.1363	0.0338	0.1372	-0.0516
17	121-33-5	Vanillin ex Eugenol, natürlich Halal Kosher, BG	19.4409	0.4368	38	0.042335	0.0096	-0.0838	0.1644	0.2204
18	16409-45-3	Menthylacetat, D,L-								

```
18.1312     0.0460              4          0.000215      -0.1996       0.050
3      -0.1068      -0.1863
   19    78-70-6       Linalool ex HO, natürlich Kosher Halal BG
17.9067     0.1264             11          0.001462       0.0995       0.135
3      -0.1975       0.0684
   20  3658-77-3                                  Furaneol Halal
17.6233     0.7931             69          0.091137       0.0007       0.102
4       0.0778       0.1003
```

```
Exported: pca_v2_noOT_top20_ingredients_global.csv / .xlsx
```




```
Saved: pca_v2_noOT_top20_global_bar.png
```

---

14 · Top 20 Ingredients per Recipe & Globally – Full Original Dataset (all 129 Recipes)

This section repeats the Top-20 analysis on the **complete original dataset** (all 129 recipes, before any outlier removal). A fresh PCA is fitted on the full dataset so that contributions are computed in the original full-data PCA space — comparable to iteration 1 of the iterative process.

In [29]:

```python
# ── Fit PCA on full original dataset (all 129 recipes) ──────────────
recipes_orig   = list(pivot_oav.index)          # all 129
X_oav_orig     = pivot_oav.values.copy()         # 129 × n_cas

scaler_orig    = StandardScaler()
X_orig_sc      = scaler_orig.fit_transform(X_oav_orig)

n_comp_orig    = min(10, X_orig_sc.shape[1], X_orig_sc.shape[0] - 1)
pca_orig       = PCA(n_components=n_comp_orig, random_state=42)
scores_orig    = pca_orig.fit_transform(X_orig_sc)
ev_orig        = pca_orig.explained_variance_ratio_ * 100
loadings_orig  = pca_orig.components_

print(f'Original dataset PCA: {len(recipes_orig)} recipes × {X_oav_orig.shap
print(f'PC1={ev_orig[0]:.1f}%  PC2={ev_orig[1]:.1f}%  PC3={ev_orig[2]:.1f}%

# Contribution tensor for all 129 recipes
N_PC_ORIG  = 4
```

```
contrib_orig = X_orig_sc[:, np.newaxis, :] * loadings_orig[np.newaxis, :N_PC

recipe_idx_orig = {r: idx for idx, r in enumerate(recipes_orig)}
print(f'Contribution tensor shape: {contrib_orig.shape}')
```

```
Original dataset PCA: 129 recipes × 225 CAS
PC1=5.2%  PC2=4.8%  PC3=4.0%  (cumulative 3: 14.0%)
Contribution tensor shape: (129, 4, 225)
```

In [30]:
```
# ── Top 20 ingredients per recipe – all 129 original recipes ──────────
N_TOP_ORIG         = 20
total_abs_orig     = np.abs(contrib_orig).sum(axis=1)  # (129, n_cas)
rows_orig          = []

for r_idx, rez in enumerate(recipes_orig):
    present_mask = pivot_oav.iloc[r_idx].values > 0
    present_idx  = np.where(present_mask)[0]
    if len(present_idx) == 0:
        continue
    contribs_pres = total_abs_orig[r_idx][present_idx]
    top_k         = min(N_TOP_ORIG, len(present_idx))
    top_local     = np.argsort(contribs_pres)[::-1][:top_k]
    top_idx_abs   = present_idx[top_local]

    outlier_flag = rez not in recipe_idx_map  # True if this recipe was remo

    for rank, i in enumerate(top_idx_abs, 1):
        cas = cas_labels[i]
        rows_orig.append({
            'Rez.-Nr.':        rez,
            'Outlier_Removed': outlier_flag,
            'Rank':            rank,
            'CAS':             cas,
            'Ingredient':      cas_name.get(cas, cas),
            'Norm_Totalmenge': round(float(pivot_oav.iloc[r_idx, i]), 6),
            'Total_Contrib':   round(float(total_abs_orig[r_idx, i]), 5),
            'PC1_Contrib':     round(float(contrib_orig[r_idx, 0, i]), 5),
            'PC2_Contrib':     round(float(contrib_orig[r_idx, 1, i]), 5),
            'PC3_Contrib':     round(float(contrib_orig[r_idx, 2, i]), 5),
            'PC4_Contrib':     round(float(contrib_orig[r_idx, 3, i]), 5),
        })

top20_orig_df = pd.DataFrame(rows_orig)
n_rec_covered = top20_orig_df['Rez.-Nr.'].nunique()
n_flagged     = top20_orig_df[top20_orig_df['Outlier_Removed']]['Rez.-Nr.'].
print(f'Shape: {top20_orig_df.shape}')
print(f'Recipes covered: {n_rec_covered} (of which {n_flagged} were removed
print(f'All Norm_Totalmenge > 0: {(top20_orig_df["Norm_Totalmenge"] > 0).all
print()
print(top20_orig_df.head(20).to_string(index=False))

top20_orig_df.to_csv(OUT_DIR / 'pca_v2_noOT_top20_per_recipe_all129.csv', ir
top20_orig_df.to_excel(OUT_DIR / 'pca_v2_noOT_top20_per_recipe_all129.xlsx',
print('\nExported: pca_v2_noOT_top20_per_recipe_all129.csv / .xlsx')
```

Shape: (2344, 11)
Recipes covered: 129 (of which 42 were removed as outliers in iterative PCA)
All Norm_Totalmenge > 0: False

Rez.-Nr.	Outlier_Removed	Rank	CAS	Ingredient	Norm_Totalmenge	Total_Contrib	PC1_Contrib	PC2_Contrib	PC3_Contrib	PC4_Contrib
185.028	True	1	97-62-1	Ethylisobutyrat Halal	0.003501	1.51250	0.56216	0.88338	0.05006	0.01690
185.028	True	2	6728-26-3	trans-2-Hexenal Kosher Halal	0.003889	0.75796	0.00821	-0.04129	-0.18458	0.52388
185.028	True	3	78-70-6	Linalool ex HO, natürlich Kosher Halal BG	0.005445	0.60283	0.22306	-0.19366	0.16129	0.02482
185.028	True	4	27625-35-0	Isoamyl-2-methylbutyrat, natürlich	0.001945	0.49667	0.00764	-0.02522	-0.12365	0.34016
185.028	True	5	3025-30-7	Ethyl-trans,cis-2,4-decadienoat, natürlich Halal	0.003112	0.41504	0.03280	0.04268	-0.09061	-0.24894
185.028	True	6	140-10-3	Zimtsäure, natürlich	0.000324	0.38061	0.05540	0.05144	-0.10011	-0.17366
185.028	True	7	23696-85-7	beta-Damascenon, natürlich Kosher Halal	0.000138	0.38061	0.05540	0.05144	-0.10011	-0.17366
185.028	True	8	75-07-0	Acetaldehyd Halal Kosher	0.125631	0.38004	0.01143	0.03413	-0.08976	-0.24471
185.028	True	9	10032-15-2	Hexyl-2-methylbutyrat Halal Kosher	0.000778	0.35491	-0.03135	-0.00632	-0.12912	0.18813
185.028	True	10	705-86-2	delta-Decalacton Halal Kosher	0.001719	0.33793	-0.00395	0.04459	-0.07078	-0.21861
185.028	True	11	503-74-2	Isovaleriansäure Halal	0.000622	0.28028	0.09086	-0.09375	-0.04343	-0.05224
185.028	True	12	79-31-2	Isobuttersäure, natürlich Halal Kosher	0.001945	0.27478	0.02219	-0.02392	-0.02264	-0.20603
185.028	True	13	64-19-7	Essigsäure Halal Kosher	0.222257	0.26997	-0.04683	0.05308	-0.02379	-0.14626
185.028	True	14	97-64-3	Ethyllactat Halal	0.126020	0.24723	-0.02415	0.00284	-0.04343	-0.17681
185.028	True	15	928-96-1	cis-3-Hexen-1-ol Halal Kosher	0.209799	0.22227	0.01986	-0.02145	-0.08272	0.09824
185.028	True	16	61931-81-5	cis-3-Hexenyllactat, natürlich Halal	0.002723	0.19918	0.01330	0.02389	-0.03214	-0.12985
185.028	True	17	108-64-5	Ethylisovale						

```
rat Kosher Halal         0.000389        0.16081     -0.02201     -0.04505
0.02623     -0.06752
 185.028             True    18     80-71-7                          Cycloten,
natürlich Halal         0.001556        0.15134      0.04879     -0.03287
-0.01866     -0.05102
 185.028             True    19   7212-44-4
Nerolidol          0.001556        0.12778      0.01208      0.02617      0.0
3456     -0.05498
 185.028             True    20    106-70-7
Methylhexanoat          0.000445        0.10261     -0.00434     -0.00555
0.05741      0.03530

Exported: pca_v2_noOT_top20_per_recipe_all129.csv / .xlsx
```

In [31]:

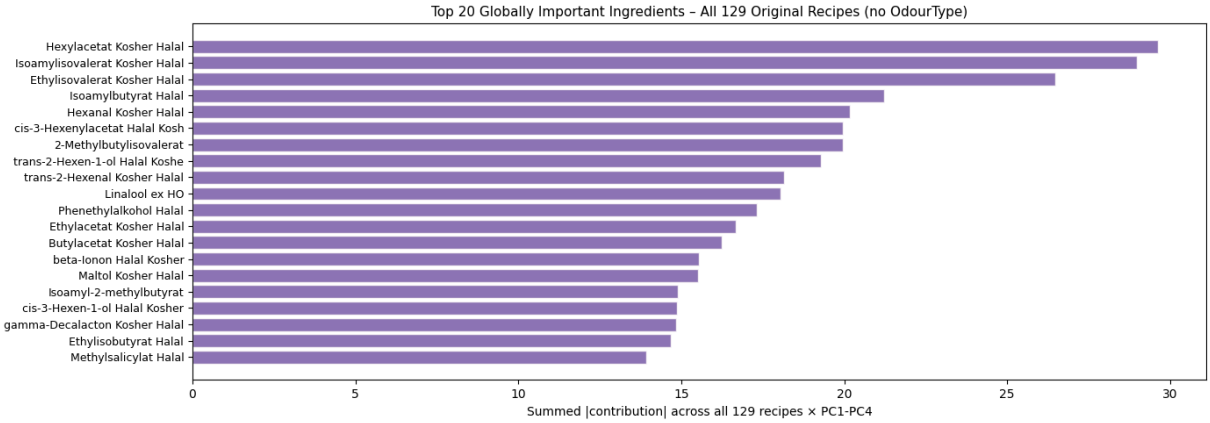
```python
# ── Top 20 ingredients globally — all 129 original recipes ──────────────
global_imp_orig     = np.abs(contrib_orig).sum(axis=(0, 1))  # (n_cas,)
top20_g_orig_idx    = np.argsort(global_imp_orig)[::-1][:20]

global_orig_rows = []
for rank, i in enumerate(top20_g_orig_idx, 1):
    cas = cas_labels[i]
    col = pivot_oav.iloc[:, i]
    global_orig_rows.append({
        'Rank':              rank,
        'CAS':               cas,
        'Ingredient':        cas_name.get(cas, cas),
        'Global_Importance': round(float(global_imp_orig[i]), 4),
        'Frequency':         round(float((col > 0).mean()), 4),
        'Recipes_Count':     int((col > 0).sum()),
        'Avg_Norm_Present':  round(float(col[col > 0].mean() if (col > 0).ar
        'PC1_Loading':       round(float(loadings_orig[0, i]), 4),
        'PC2_Loading':       round(float(loadings_orig[1, i]), 4),
        'PC3_Loading':       round(float(loadings_orig[2, i]), 4),
        'PC4_Loading':       round(float(loadings_orig[3, i]), 4),
    })

top20_g_orig_df = pd.DataFrame(global_orig_rows)
print('=== Top 20 Ingredients Globally — All 129 Original Recipes ===')
print(top20_g_orig_df.to_string(index=False))

top20_g_orig_df.to_csv(OUT_DIR / 'pca_v2_noOT_top20_ingredients_global_all12
top20_g_orig_df.to_excel(OUT_DIR / 'pca_v2_noOT_top20_ingredients_global_all
print('\nExported: pca_v2_noOT_top20_ingredients_global_all129.csv / .xlsx')

names_short_orig = [cas_name.get(cas_labels[i], cas_labels[i]).split(',')[0]
                    for i in top20_g_orig_idx]
fig, ax = plt.subplots(figsize=(14, 5))
ax.barh(range(20), global_imp_orig[top20_g_orig_idx[::-1]],
        color='#7B5EA7', alpha=0.85, edgecolor='white')
ax.set_yticks(range(20))
ax.set_yticklabels(names_short_orig[::-1], fontsize=9)
ax.set_xlabel('Summed |contribution| across all 129 recipes × PC1–PC4', font
ax.set_title(
    'Top 20 Globally Important Ingredients — All 129 Original Recipes (no Oc
    fontsize=11
)
```

```
plt.tight_layout()
fig.savefig(OUT_DIR / 'pca_v2_noOT_top20_global_bar_all129.png', dpi=150, bb
plt.show()
print('Saved: pca_v2_noOT_top20_global_bar_all129.png')
```

=== Top 20 Ingredients Globally – All 129 Original Recipes ===

Rank	CAS	Ingredient	Global_Importance	Frequency	Recipes_Count	Avg_Norm_Present	PC1_Loading	PC2_Loading	PC3_Loading	PC4_Loading
1	142-92-7	Hexylacetat Kosher Halal	29.6281	0.5116	66	0.008638	0.0048	-0.0228	-0.0971	0.2674
2	659-70-1	Isoamylisovalerat Kosher Halal	28.9910	0.3798	49	0.011154	-0.0088	-0.0165	-0.0890	0.2448
3	108-64-5	Ethylisovalerat Kosher Halal	26.4767	0.5814	75	0.015013	0.0431	0.0881	-0.0513	0.1321
4	106-27-4	Isoamylbutyrat Halal	21.2149	0.1938	25	0.009425	-0.0015	-0.0174	-0.0871	0.2380
5	66-25-1	Hexanal Kosher Halal	20.1774	0.1860	24	0.001571	-0.0184	-0.0222	0.2745	0.0888
6	3681-71-8	cis-3-Hexenylacetat Halal Kosher	19.9602	0.7442	96	0.011774	0.0432	-0.0010	-0.0593	0.1178
7	2445-77-4	2-Methylbutylisovalerat	19.9592	0.1240	16	0.003091	0.0015	-0.0160	-0.0962	0.2602
8	2305-21-7	trans-2-Hexen-1-ol Halal Kosher	19.2800	0.1705	22	0.011864	-0.0019	-0.0186	-0.0894	0.2469
9	6728-26-3	trans-2-Hexenal Kosher Halal	18.1429	0.1783	23	0.002183	0.0037	-0.0188	-0.0841	0.2387
10	78-70-6	Linalool ex HO, natürlich Kosher Halal BG	18.0587	0.2636	34	0.002670	0.0901	-0.0783	0.0652	0.0100
11	60-12-8	Phenethylalkohol Halal	17.3115	0.0465	6	0.020753	0.2228	-0.1644	-0.0370	0.0922
12	141-78-6	Ethylacetat Kosher Halal	16.6806	0.6124	79	0.033561	0.0906	-0.0783	0.0383	0.0094
13	123-86-4	Butylacetat Kosher Halal	16.2507	0.1008	13	0.002833	-0.0099	-0.0150	0.2966	0.1010
14	14901-07-6	beta-Ionon Halal Kosher	15.5498	0.0853	11	0.001304	0.1060	0.1682	0.0271	-0.0078
15	118-71-8	Maltol Kosher Halal	15.5115	0.5116	66	0.046309	0.0218	0.0431	-0.0386	0.0761
16	27625-35-0	Isoamyl-2-methylbutyrat, natürlich	14.8970	0.0388	5	0.005413	0.0054	-0.0177	-0.0866	0.2383
17	928-96-1	cis-3-Hexen-1-ol Halal Kosher	14.8794	0.8992	116	0.088893	0.0136	-0.0147	-0.0566	0.0672
18	706-14-9	gamma-Decalacton Kosher Halal	14.8520							

```
9     0.9302            120          0.038739       0.0053      -0.0376
-0.0388       0.0937
   19    97-62-1                     Ethylisobutyrat Halal           14.690
7     0.1163             15          0.001173       0.1418       0.2229
0.0126       0.0043
   20   119-36-8                     Methylsalicylat Halal           13.916
7     0.0853             11          0.005185       0.0071      -0.0200
-0.0897       0.2663

Exported: pca_v2_noOT_top20_ingredients_global_all129.csv / .xlsx
```




```
Saved: pca_v2_noOT_top20_global_bar_all129.png
```

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14 · Key Findings (v2 - No Threshold, No OdourType)

Data Processing

- Ignore list applied (CAS-based masking) before any analysis.
- Totalmenge normalised per recipe (sum = 1) — no log transform, no OAV threshold.
- OdourType information intentionally excluded to let the chemical fingerprint speak alone.

Outlier Removal

- Iterative 2D Mahalanobis outlier detection in PC1-PC2 space (threshold = 3σ).
- Process repeats until no outliers remain or 20 iterations are exhausted.
- See Section 8 for the full outlier log.

Interpretation

- PC1 captures the dominant compositional axis (major aroma directions).
- PC2 captures secondary compositional variation.
- Clusters in this space represent recipes with similar CAS-level fingerprints, independent of any odour-type classification.
- The biplot arrows show which specific ingredients drive cluster separation.

Summary

	Value
Recipes analysed (start)	—
Recipes after outlier removal	—
CAS features	—
Outlier threshold	3σ Mahalanobis (2D PC1-PC2)
OdourType	Not used
Outputs	`outputs/pca_v2_noOT_*`